5 Dirichlet averages

5.1 Introduction

A Dirichlet average of a function denotes a certain kind of integral average with respect to a Dirichlet measure. Many of the most important special functions can be represented as Dirichlet averages of the functions e^x and x^t, and there are significant advantages in defining them this way instead of by hypergeometric power series. Properties of higher functions are illuminated by analogy with familiar properties of the exponential and power functions, and some complicated transformations are replaced by a permutation symmetry that is hidden from the conventional point of view. The theory of Dirichlet averages is not restricted to functions of hypergeometric type since any function that is analytic or even integrable can be averaged with respect to a Dirichlet measure. If the function is twice continuously differentiable, its Dirichlet average satisfies one or more linear second-order partial differential equations that are characteristic of the averaging process (Section 5.4) and are related to some of the principal differential equations of mathematical physics.

Unrestricted integral averages with respect to Dirichlet measure were discussed briefly in Section 4.4. In the simple case in which the function being averaged depends only on a sum of integration variables, the multiple integral can be reduced to a single integral by (4.4-12). That which we call a Dirichlet average is between these two extremes but includes the second as a special case: the function being averaged depends only on a linear function of the integration variables. The coefficients of the linear function will be regarded as independent variables, and we shall consider in this chapter how the Dirichlet average depends on these variables and on the parameters of the Dirichlet measure.

As a model for multiple integrals we start with the single integral

$$F(\beta, \beta'; x, y) = \int_0^1 f[ux + (1-u)y] \frac{u^{\beta-1}(1-u)^{\beta'-1}}{\mathrm{B}(\beta, \beta')} \, du, \qquad \beta, \beta' \in \mathbb{C}_>. \quad (1)$$

The linear function $u \mapsto ux + (1-u)y$ maps the unit interval of integration onto the interval in $\mathbb{R}$, or more generally the line segment in $\mathbb{C}$, with endpoints x and y. The function f must of course be such that the integral has a meaning. If β and β' are real and positive, F is an integral average of f over the line segment.

By analogy with (1) let $z = (z_1, \ldots, z_k) \in \mathbb{C}^k$, $k \geq 2$, and consider the multiple integral

$$F(b, z) = \int_E f(u \cdot z) \, d\mu_b(u), \quad (2)$$

where μ_b is a Dirichlet measure on the standard simplex E in $\mathbb{R}^{k-1}$ and

$$u \cdot z = \sum_{i=1}^{k} u_i z_i, \qquad u_k = 1 - u_1 - \cdots - u_{k-1}. \quad (3)$$

The quantity $u \cdot z$ is both a convex combination of $z_1, \ldots, z_k$ and a linear function of the independent variables $u_1, \ldots, u_{k-1}$. From the latter viewpoint $u \cdot z$ maps the standard simplex E onto a set called the convex hull of $\{z_1, \ldots, z_k\}$, which we denote by $\mathrm{con}(z)$ and define formally as follows.

DEFINITION 5.1-1 Let $z = (z_1, \ldots, z_k) \in \mathbb{C}^k$, $k \geq 2$, and let E be the standard simplex in $\mathbb{R}^{k-1}$. The convex hull of $\{z_1, \ldots, z_k\}$ is defined to be

$$\mathrm{con}(z) = \left\{ \zeta \in \mathbb{C} : \zeta = \sum_{i=1}^{k} u_i z_i, \, (u_1, \ldots, u_{k-1}) \in E, \, \sum_{i=1}^{k} u_i = 1 \right\}. \quad (4)$$

The convex hull of $\{z_1, z_2\}$ is the line segment with endpoints z_1 and z_2. In general $\mathrm{con}(z)$ is a polygon (including its interior) in $\mathbb{C}$ with at most k vertices (see Fig. 5.1-1). There may be fewer than k vertices (e.g., if $z_1, \ldots, z_k$ are all real and $k > 2$), but every vertex is one of the z_i.

We tacitly assumed in (2) that f is defined on $\mathrm{con}(z)$ and the integral has a meaning. It suffices that f be defined and measurable on a convex set $\Omega \subset \mathbb{C}$ containing $z_1, \ldots, z_k$, for $\mathrm{con}(z)$ is the smallest convex set containing $z_1, \ldots, z_k$ (i.e., the intersection of all such convex sets) and thus $\mathrm{con}(z) \subset \Omega$.

Since $\mu_b(E) = 1$, we can think of $F(b, z)$ as an integral average of f over $\mathrm{con}(z)$. Strictly speaking, it is an average only if the b parameters are real

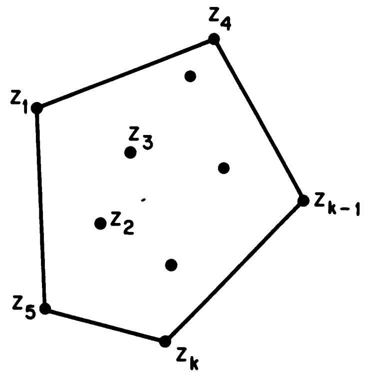

Figure 5.1-1 If $z \in \mathbb{C}^k$ the convex hull of $\{z_1, \ldots, z_k\}$ is a polygon (including its interior) in the complex plane with at most k vertices.

and positive, but we shall call F a Dirichlet average of f even when the parameters are complex. Dirichlet averaging is easily extended to a function f defined on any topological vector space whose scalars include the real numbers, but we shall have no occasion to do so.

If f is an analytic function of one complex variable, then F is an analytic function of the several complex variables $z_1, \ldots, z_k$. This is to be expected because an integral average is a kind of sum, and a sum (with qualifications) of analytic functions is analytic. The most elementary analytic functions, e^x and x^t, will be discussed briefly in Sections 5.8–5.9 and in more detail later. It is remarkable that Dirichlet averages of these two elementary functions yield so many of the well-known special functions of applied mathematics.

5.2 The averaging process

After formally defining a Dirichlet average, we shall establish several fundamental properties that are independent of the function being averaged. The most important of these is the property of permutation symmetry (Theorem 5.2-3).

DEFINITION 5.2-1 Let Ω be a convex set in $\mathbb{C}$, let $z = (z_1, \ldots, z_k) \in \Omega^k$, $k \geq 2$, and let $u \cdot z$ be a convex combination of $z_1, \ldots, z_k$. Let f be a measurable function on Ω, and let μ_b be a Dirichlet measure on the standard simplex E in $\mathbb{R}^{k-1}$. Define

$$F(b, z) = \int_E f(u \cdot z)\, d\mu_b(u). \tag{1}$$

We shall call F the Dirichlet average of f with variables $z = (z_1, \ldots, z_k)$ and parameters $b = (b_1, \ldots, b_k)$. If $k = 1$, we define $F(b, z) = f(z)$.

Remark If f is continuous on Ω, the integral exists and has a finite value both as a possibly improper Riemann integral and as a Lebesgue integral because $f(u \cdot z)$ is continuous and therefore bounded on the compact set E. Thus the absolute value of the integral does not exceed a finite constant times $|\mu_b|(E)$, which is finite for every Dirichlet measure by (4.4-4). Continuity of f is sufficient but far from necessary for the existence of F,

If $z_1 = \cdots = z_k = \zeta$, then $u \cdot z = \zeta$. Since $\mu_b(E) = 1$,

$$F(b_1, \ldots, b_k; \zeta, \ldots, \zeta) = f(\zeta). \tag{2}$$

We say that f is the restriction of F to the diagonal of Ω^k.

THEOREM 5.2-2 Let F be defined by Definition 5.2-1, and assume $f\colon \Omega \to \mathbb{C}$ is continuous. Then $F(b, z)$ is holomorphic in b on $\mathbb{C}_>{}^k$ for every fixed $z \in \Omega^k$.

Proof The case $k = 1$ is trivial. If $k \geq 2$, the function $[(u_1, \ldots, u_{k-1}) \mapsto f(u \cdot z)]$ is continuous on E for every $z \in \Omega^k$, and the conclusion of the theorem follows from Theorem 4.4-6. ▯

THEOREM 5.2-3 Let F be defined by Definition 5.2-1, and assume f is continuous on Ω. Then $F(b, z) = F(b_1, \ldots, b_k; z_1, \ldots, z_k)$ is symmetric in the indices $1, \ldots, k$.

Proof In (1) apply a permutation of the indices to $f(u \cdot z) = f(u_1 z_1 + \cdots + u_k z_k)$. On one hand, $f(u \cdot z)$ has the same value as before and consequently so does $F(b, z)$. On the other hand, the variables $z_1, \ldots, z_k$ of F are permuted, and the permutation of $u_1, \ldots, u_k$ induces the same permutation of the parameters $b_1, \ldots, b_k$ of F according to Theorem 4.4-2. Hence simultaneous permutation of variables and parameters leaves the value of F unchanged. ▯

Remark The discussion of symmetry in the last paragraph of Section 4.3 applies equally well here if we think of z_i and b_i as being attached to the same vertex of E. We can rewrite (1) as an integral over the unit cube in $\mathbb{R}^k$ by inserting a Dirac δ function as in (4.3-11).

THEOREM 5.2-4 Let F be defined by Definition 5.2-1, and assume $f\colon \Omega \to \mathbb{C}$ is continuous. Let p and k be integers such that $0 \leq p < k$, and define $\beta = b_{p+1} + \cdots + b_k$. If $z_i = \zeta$ for $i = p + 1, \ldots, k$, then

$$F(b_1, \ldots, b_k; z_1, \ldots, z_k) = F(b_1, \ldots, b_p, \beta; z_1, \ldots, z_p, \zeta), \tag{3}$$

where the left and right sides are Dirichlet averages of f in k and $p + 1$ variables, respectively. If $p = 0$, the right side is understood to be $f(\zeta)$.

Remark The theorem states that equal arguments can be replaced by a single argument if the corresponding b parameters are replaced by their sum. According to the symmetry property established in Theorem 5.2-2, the indices of the equal arguments (chosen to be $p + 1, \ldots, k$ for convenience) are immaterial.

Proof Since the case $p = 0$ is (2), we may assume $1 \leq p < k$. Because $u \cdot z = u_1 z_1 + \cdots + u_p z_p + (u_{p+1} + \cdots + u_k)\zeta$, we may set $k = p + q$ and apply Theorem 4.4-3. □

EXAMPLE 5.2-5 The convenience of the notation and the use of the theorem are illustrated by reconsidering the reducible integrals in Corollaries 4.4-4 and 4.4-5. We denote the Dirichlet average of φ by Φ. The left side of (4.4-12) is $\Phi(b_1, \ldots, b_{p+q}; 1, \ldots, 1, 0, \ldots, 0)$, with p unit arguments and q zero arguments. By two applications of Theorem 5.2-4, we may replace each set of equal arguments by a single argument and obtain

$$\Phi(b_1 + \cdots + b_p, b_{p+1} + \cdots + b_{p+q}; 1, 0),$$

which indeed equals the single integral on the right side of (4.4-12).

Dirichlet's integral on the left side of (4.4-14) is

$$\begin{aligned} &\mathrm{B}(b_1, \ldots, b_p, 1)\Phi(b_1, \ldots, b_p, 1; 1, \ldots, 1, 0) \\ &\quad = \mathrm{B}(b_1, \ldots, b_p, 1)\Phi(b_1 + \cdots + b_p, 1; 1, 0) \\ &\quad = \frac{\mathrm{B}(b_1, \ldots, b_p, 1)}{\mathrm{B}(b_1 + \cdots + b_p, 1)} \int_0^1 \varphi(u) u^{b_1 + \cdots + b_p - 1}\, du \\ &\quad = \mathrm{B}(b_1, \ldots, b_p) \int_0^1 \varphi(u) u^{b_1 + \cdots + b_p - 1}\, du. \end{aligned}$$

The last expression is the right side of (4.4-14).

We establish next the very useful result that the Dirichlet average of $f(\alpha x + \lambda)$ is $F(b, \alpha z + \lambda)$, where F is defined by (1), α and λ are constants, and $\alpha z + \lambda = (\alpha z_1 + \lambda, \ldots, \alpha z_k + \lambda)$.

THEOREM 5.2-6 Let $\alpha, \lambda \in \mathbb{C}$, and let Ω be a convex set in $\mathbb{C}$. Let f be a measurable function on Ω, and define a function g by

$$g(x) = f(\alpha x + \lambda), \qquad \alpha x + \lambda \in \Omega.$$

If $\alpha z + \lambda = (\alpha z_1 + \lambda, \ldots, \alpha z_k + \lambda) \in \Omega^k$, then the Dirichlet average of g is

$$G(b, z) = F(b, \alpha z + \lambda).$$

Proof The case $k = 1$ is trivial. If $k \geq 2$, then

$$F(b, \alpha z + \lambda) = \int_E f[u \cdot (\alpha z + \lambda)]\, d\mu_b(u).$$

Now

$$u \cdot (\alpha z + \lambda) = \alpha(u_1 z_1 + \cdots + u_k z_k) + (u_1 + \cdots + u_k)\lambda = \alpha(u \cdot z) + \lambda$$

and

$$f[u \cdot (\alpha z + \lambda)] = f[\alpha(u \cdot z) + \lambda] = g(u \cdot z)$$

Hence

$$F(b, \alpha z + \lambda) = \int_E g(u \cdot z)\, d\mu_b(u) = G(b,z). \quad \square$$

5.3 Averages of derivatives

We ask now whether $F(b, z)$ has derivatives with respect to $z_1, \ldots, z_k$ and whether differentiations can be taken under the integral sign. A small amount of drudgery will establish (for real variables) that if f is n times continuously differentiable, so is F, and (for complex variables) that if f is analytic, F is also analytic. The Dirichlet average of a derivative of f has a simple relation to the partial derivatives of F.

If Ω is an open set in $\mathbb{R}$, let $C_n(\Omega)$ be the class of real-valued functions which are n times continuously differentiable on Ω. In particular, $C_0(\Omega)$ is the class of continuous functions. Note that $C_n(\Omega) \supset C_p(\Omega)$ if $n \leq p$. If $f \in C_n(\Omega)$, we denote its derivatives by $f^{(0)} = f, f^{(1)} = f', f^{(2)} = f'', \ldots, f^{(n)}$. The same notation will be used for complex derivatives if f is a holomorphic function on an open set Ω in $\mathbb{C}$.

DEFINITION 5.3-1 Let $n \in \mathbb{N}$ and assume that either

(i) Ω is an open interval in $\mathbb{R}$ and $f \in C_n(\Omega)$, or
(ii) Ω is a convex open set in $\mathbb{C}$ and f is holomorphic on Ω.

If $k \geq 2$, let μ_b be a Dirichlet measure on the standard simplex E in $\mathbb{R}^{k-1}$. For every $z \in \Omega^k$ define

$$F^{(n)}(b, z) = \int_E f^{(n)}(u \cdot z)\, d\mu_b(u). \tag{1}$$

Alternative notations are $F = F^{(0)}$, $F' = F^{(1)}$, $F'' = F^{(2)}$. If $k = 1$, define $F^{(n)} = f^{(n)}$.

We write $[z \mapsto F(b, z)]$ to denote $F(b, z)$ as a function of z for fixed b (see Appendix B). We use the abbreviations $D_i = \partial/\partial z_i$ and $D = \sum_{i=1}^{k} D_i$. If $g(z)$ has continuous partial derivatives of order n on Ω^k, we write $g \in C_n(\Omega^k)$.

THEOREM 5.3-2 Let Ω be an open interval in $\mathbb{R}$, $n \in \mathbb{N}$, and $f \in C_n(\Omega)$. Let $z \in \Omega^k$ and define $u \cdot z$ by (5.1-3). Let $m \in \mathbb{N}^k$ and assume $\sum_{i=1}^{k} m_i = n$. Then $[z \mapsto F(b, z)] \in C_n(\Omega^k)$ and

$$D_1^{m_1} \cdots D_k^{m_k} F(b, z) = \int_E u_1^{m_1} \cdots u_k^{m_k} f^{(n)}(u \cdot z)\, d\mu_b(u), \qquad k \geq 2. \tag{2}$$

Furthermore,

$$D^n F = F^{(n)}, \tag{3}$$

where $D = \sum_{i=1}^{k} D_i$. More generally, if α, β, γ are complex constants, define

$$\delta = \alpha + \beta \frac{d}{dx} + \gamma x \frac{d}{dx}, \qquad \Delta = \alpha + \beta \sum_{i=1}^{k} D_i + \gamma \sum_{i=1}^{k} z_i D_i, \tag{4}$$

$$g_n(x) = \delta^n f(x), \qquad G_n(b, z) = \Delta^n F(b, z).$$

Then the Dirichlet average of g_n is G_n.

Remark Equation (2) asserts that F can be differentiated under the integral sign. If $\alpha = \gamma = 0$ and $\beta = 1$, then $g_n = f^{(n)}$ and $G_n = D^n F$ in agreement with (1) and (3).

Proof Since the case $k = 1$ is trivial, assume $k \geq 2$. In Theorem B.3 choose $\mu = \mu_b$, $T = E$, $X = \Omega^k$, and $\|m\| = n$. Let $u' = (u_1, \ldots, u_{k-1})$ and verify the three hypotheses of that theorem as follows.

(i) $[z \mapsto f(u \cdot z)] \in C_n(\Omega^k)$ because $f \in C_n(\Omega)$.

(ii) If $\|m\| = n$, $[u' \mapsto D_z^m f(u \cdot z)]$ is continuous on the compact set E. Hence the integrand of (2) is bounded and $|\mu_b|$-integrable.

(iii) If K is a compact set in Ω^k and $\|m\| = n$, the function $[(u', z) \mapsto D_z^m f(u \cdot z)]$ is continuous on the compact set $E \times K$ and so is bounded. We may choose g_K^m to be a constant, which is $|\mu_b|$-integrable by (4.4-4).

We conclude from Theorem B.3 that $[z \mapsto F(b, z)] \in C_n(\Omega^k)$ and that (2) is true. Since $\sum_{i=1}^{k} u_i = 1$, (2) implies (3) as follows:

$$\left(\sum_{i=1}^{k} D_i\right)^n F(b, z) = \int_E \left(\sum_{i=1}^{k} u_i\right)^n f^{(n)}(u \cdot z)\, d\mu_b(u)$$

$$= \int_E f^{(n)}(u \cdot z)\, d\mu_b(u) = F^{(n)}(b, z).$$

The Dirichlet average of $g_0 = f$ is $G_0 = F$. We make the inductive assumption that the Dirichlet average of g_{n-1} is G_{n-1}. Then

$$G_n(b, z) = \Delta G_{n-1}(b, z) = \Delta \int_E g_{n-1}(u \cdot z)\, d\mu_b(u).$$

We use (2) and find

$$G_n(b, z) = \int_E [\alpha g_{n-1}(u \cdot z) + \beta g'_{n-1}(u \cdot z) + \gamma(u \cdot z) g'_{n-1}(u \cdot z)]\, d\mu_b(u)$$

$$= \int_E \delta g_{n-1}(u \cdot z)\, d\mu_b(u) = \int_E g_n(u \cdot z)\, d\mu_b(u). \qquad \square$$

THEOREM 5.3-3 Let Ω be a convex open set in $\mathbb{C}$ and let $f: \Omega \to \mathbb{C}$ be holomorphic. Then the Dirichlet average $F(b, z)$ is holomorphic on $\mathbb{C}_>^k \times \Omega^k$, and (2)–(4) are valid for every $n \in \mathbb{N}$.

Proof Since the case $k = 1$ is trivial, assume $k \geq 2$. By Theorem 5.2-6, $[b \mapsto F(b, z)]$ is holomorphic on $\mathbb{C}_>{}^k$ for every fixed $z \in \Omega^k$. Since a function is holomorphic in several variables jointly if it is holomorphic in each variable separately, it suffices to prove that $[z \mapsto F(b, z)]$ is holomorphic on Ω^k for each fixed $b \in \mathbb{C}_>{}^k$. In Theorem B.4 choose $\mu = \mu_b$, $T = E$, and $Z = \Omega^k$. Let $u' = (u_1, \ldots, u_{k-1})$ and verify the three hypotheses of that theorem as follows.

(i) $[z \mapsto f(u \cdot z)]$ is holomorphic on Ω^k because f is holomorphic on Ω.

(ii) $[u' \mapsto f(u \cdot z)]$ is continuous on the compact set E and is therefore bounded and $|\mu_b|$-integrable.

(iii) If K is a compact set in Ω^k, the function $[(u', z) \mapsto f(u \cdot z)]$ is continuous on the compact set $E \times K$ and so is bounded. We may choose g_K to be a constant.

We conclude from Theorem B.4 that $F(b, z)$ is holomorphic in z on Ω^k and that all its partial derivatives (which also are holomorphic) can be computed by differentiating under the integral sign, as in (2). Equations (3) and (4) follow from (2) as in the proof of Theorem 5.3-2. □

5.4 Euler–Poisson differential equations

The results of the last section on differentiating $F(b, z)$ under the integral sign will be used now to derive a system of partial differential equations characteristic of the Dirichlet averaging process. Before discussing k variables we consider two variables.

If v denotes $F(\beta, \beta'; x, y)$, then

$$\mathrm{B}(\beta, \beta')v = \int_0^1 f[ux + (1 - u)y]u^{\beta-1}(1 - u)^{\beta'-1}\, du.$$

Assuming f to be twice continuously differentiable, we use (5.3-2) to obtain

$$\begin{aligned}\mathrm{B}(\beta, \beta')(x - y)\frac{\partial^2 v}{\partial x\, \partial y} &= \int_0^1 (x - y)f''[ux + (1 - u)y]u^{\beta}(1 - u)^{\beta'}\, du \\ &= \int_0^1 \left\{\frac{d}{du} f'[ux + (1 - u)y]\right\} u^{\beta}(1 - u)^{\beta'}\, du \\ &= -\int_0^1 f'[ux + (1 - u)y]\frac{d}{du}\{u^{\beta}(1 - u)^{\beta'}\}\, du.\end{aligned}$$

The last equality results from integration by parts, and the boundary terms vanish because β and β' have positive real parts. We perform the differentiation with respect to u and find

$$\begin{aligned}(x-y)\frac{\partial^2 v}{\partial x\,\partial y} &= -\int_0^1 f'[ux+(1-u)y]\{\beta(1-u)-\beta' u\}\frac{u^{\beta-1}(1-u)^{\beta'-1}}{B(\beta,\beta')}\,du\\ &= -\beta\frac{\partial v}{\partial y}+\beta'\frac{\partial v}{\partial x},\end{aligned}$$

where the last step again uses (5.3-2). Thus $v = F(\beta, \beta'; x, y)$ is a solution of

$$(x-y)\frac{\partial^2 v}{\partial x\,\partial y}+\beta\frac{\partial v}{\partial y}-\beta'\frac{\partial v}{\partial x}=0. \tag{1}$$

This equation was called the equation of Euler and Poisson by Darboux (1894, Vol. 2, pp. 54–70), who observed that it is related to the hypergeometric equation. Appell (1926, p. 53) encountered it in connection with a double hypergeometric series.

We shall show now that a Dirichlet average in any number of variables satisfies a system of Euler–Poisson equations. Such systems are discussed briefly by Darboux (1894, Vol. 4, pp. 280–281) and Appell (1926, p. 119).

THEOREM 5.4-1 Assume that either

(i) Ω is an open interval of the real line and $f\colon \Omega \to \mathbb{C}$ is twice continuously differentiable, or

(ii) Ω is a convex open set in the complex plane and $f\colon \Omega \to \mathbb{C}$ is holomorphic.

Let $F(b, z)$ be defined on $\mathbb{C}_>{}^k \times \Omega^k$, $k \geq 2$, by Definition 5.2-1, and write $D_i = \partial/\partial z_i$. Then $v = F(b, z)$ is a solution of the system of Euler–Poisson differential equations,

$$(z_i - z_j)D_i D_j v + b_i D_j v - b_j D_i v = 0, \qquad i, j = 1, \ldots, k. \tag{2}$$

Proof The symmetry of F (Theorem 5.2-3) implies that it is sufficient to prove the equations with $j = k$, and since the equations with $i = j$ are trivial, we may assume $i < k$. According to (5.3-2),

$$\begin{aligned}(z_i - z_k)D_i D_k v &= \int_E (z_i - z_k)D_i D_k f(u\cdot z)\,d\mu_b(u)\\ &= \int_E (z_i - z_k)f''(u\cdot z)u_i u_k\,d\mu_b(u).\end{aligned}$$

Since $u_k = 1 - u_1 - \cdots - u_{k-1}$,

$$(z_i - z_k)f''(u\cdot z) = \frac{\partial}{\partial u_i}f'(u\cdot z)$$

and thus

$$(z_i - z_k)D_i D_k v = \frac{1}{\mathrm{B}(b)} \int_E \left[\frac{\partial}{\partial u_i} f'(u \cdot z)\right] u_i u_k \prod_{r=1}^{k} u_r^{b_r - 1}\, du_1 \cdots du_{k-1}.$$

We now integrate by parts with respect to u_i, the limits of integration being determined by $u_i = 0$ and $1 - u_1 - \cdots - u_{k-1} = u_k = 0$. The boundary terms vanish because b_i and b_k have positive real parts. Hence

$$\begin{aligned}(z_i - z_k)D_i D_k v &= \frac{-1}{\mathrm{B}(b)} \int_E f'(u \cdot z) \frac{\partial}{\partial u_i}\left[u_i u_k \prod_{r=1}^{k} u_r^{b_r - 1}\right] du_1 \cdots du_{k-1} \\ &= \frac{-1}{\mathrm{B}(b)} \int_E f'(u \cdot z)(b_i u_k - b_k u_i) \prod_{r=1}^{k} u_r^{b_r - 1}\, du_1 \cdots du_{k-1} \\ &= -b_i D_k v + b_k D_i v. \quad \square\end{aligned}$$

A change of variables in (1) shows that the function ψ defined by (3) satisfies the differential equation (4):

$$\psi = F(\beta, \beta'; x + y, x - y), \tag{3}$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\beta - \beta'}{y}\frac{\partial \psi}{\partial x} = \frac{\partial^2 \psi}{\partial y^2} + \frac{\beta + \beta'}{y}\frac{\partial \psi}{\partial y}. \tag{4}$$

If $\beta = \beta'$, (4) becomes the Euler–Poisson–Darboux equation:

$$w = F(\beta, \beta; x + y, x - y), \tag{5}$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial^2 w}{\partial y^2} + \frac{2\beta}{y}\frac{\partial w}{\partial y}. \tag{6}$$

Replacing y by iy, where $i^2 = -1$, leads to the equation of generalized axially symmetric potential theory:

$$v = F(\beta, \beta; x + iy, x - iy), \tag{7}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{2\beta}{y}\frac{\partial v}{\partial y} = 0. \tag{8}$$

If x and y are real, (6) is an equation of hyperbolic type and (8) of elliptic type. Both are connected with fundamental equations of mathematical physics as follows. If

$$g = g(y), \qquad y = \left(\sum_{i=1}^{p} x_i^2\right)^{1/2}, \tag{9}$$

then

$$\sum_{i=1}^{p} \frac{\partial^2 g}{\partial x_i^2} = \frac{d^2 g}{dy^2} + \frac{p-1}{y}\frac{dg}{dy}. \tag{10}$$

In $\mathbb{R}^n$ we define the radial coordinate $r = (\sum_{i=1}^{n} x_i^2)^{1/2}$, and in (5) and (6) we set $x = t$, $y = r$, and $2\beta = n - 1$. Assuming $n \geq 2$ we find a spherically symmetric solution of the wave equation,

$$W = F\left(\frac{n-1}{2}, \frac{n-1}{2}; t + r, t - r\right), \qquad r = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}, \tag{11}$$

$$\frac{\partial^2 W}{\partial t^2} = \sum_{i=1}^{n} \frac{\partial^2 W}{\partial x_i^2}. \tag{12}$$

In (7) and (8) we choose $x = x_n$ and denote the distance from the x_n axis by $\rho = (\sum_{i=1}^{n-1} x_i^2)^{1/2}$. Setting $y = \rho$ and $2\beta = n - 2$, we find an axially symmetric solution of Laplace's equation if $n \geq 3$,

$$V = F\left(\frac{n-2}{2}, \frac{n-2}{2}; x_n + i\rho, x_n - i\rho\right), \qquad \rho = \left(\sum_{i=1}^{n-1} x_i^2\right)^{1/2}, \tag{13}$$

$$\sum_{i=1}^{n} \frac{\partial^2 V}{\partial x_i^2} = 0. \tag{14}$$

A case of particular interest is the axially symmetric potential in three dimensions,

$$V(x, y, z) = F[\tfrac{1}{2}, \tfrac{1}{2}; z + i(x^2 + y^2)^{1/2}, z - i(x^2 + y^2)^{1/2}], \tag{15}$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0. \tag{16}$$

Note that $V(0, 0, z) = f(z)$ by (5.2-2).

We substitute $\psi = y^\alpha \varphi$ in (4), where α is a constant, and find

$$\varphi = y^{-\alpha} F(\beta, \beta'; x + y, x - y), \tag{17}$$

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\beta - \beta'}{y}\frac{\partial \varphi}{\partial x} = \frac{\partial^2 \varphi}{\partial y^2} + \frac{\beta + \beta' + 2\alpha}{y}\frac{\partial \varphi}{\partial y} + \frac{\alpha(\beta + \beta' + \alpha - 1)}{y^2}\varphi. \tag{18}$$

If the value of the coefficient in the last term is given, there are two choices for α. Denote these by $\alpha = \lambda \mp \mu$, where

$$\alpha(\beta + \beta' + \alpha - 1) = \lambda^2 - \mu^2 = (\lambda + \mu)(\lambda - \mu).$$

Then $\beta + \beta' = 1 \pm 2\mu$. If we set $\beta - \beta' = 2\nu$, then

$$\varphi = y^{-\lambda \pm \mu} F(\tfrac{1}{2} \pm \mu + \nu, \tfrac{1}{2} \pm \mu - \nu; x + y, x - y), \tag{19}$$

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{2\nu}{y}\frac{\partial \varphi}{\partial x} = \frac{\partial^2 \varphi}{\partial y^2} + \frac{2\lambda + 1}{y}\frac{\partial \varphi}{\partial y} + \frac{\lambda^2 - \mu^2}{y^2}\varphi. \tag{20}$$

Equation (20) is often encountered when angular variables are separated from a partial differential equation containing the Laplacian operator. An example to be discussed in Section 5.8 is the Schrödinger equation in a Coulomb field. If $\lambda = \mu = (\beta + \beta' - 1)/2$, the last term in (20) vanishes and we recover (4). However, choice of the lower signs in (19) shows that a second solution of (4) is

$$\psi = y^{1-\beta-\beta'} F(1-\beta', 1-\beta; x+y, x-y), \tag{21}$$

provided the difference between the exponents, $\pm 2\mu = \beta + \beta' - 1$, is not an integer. At present (21) has a meaning only if $\operatorname{Re}\beta < 1$ and $\operatorname{Re}\beta' < 1$, but this condition will be relaxed in Section 6.3. Corresponding solutions of (6) and (8) can be obtained from (21) by inspection.

In addition to satisfying the Euler–Poisson equations, which are characteristic of the averaging process itself, the Dirichlet average $F(b, z)$ may obey other differential equations deriving from the properties of f. In particular, if f satisfies a linear equation with constant coefficients,

$$\sum_{n=0}^{p} a_n \frac{d^n f}{dx^n} = 0, \tag{22}$$

then (5.3-3) implies that

$$\sum_{n=0}^{p} a_n D^n F = 0, \tag{23}$$

where $D = \sum_{i=1}^{k} D_i$.

5.5 Newton–Taylor series with remainder

A particularly interesting case of the Dirichlet average is the unweighted average. This is the case $b_1 = \cdots = b_k = 1$, $k \geq 2$, in which Dirichlet measure reduces to Lebesgue measure:

$$d\mu_b(u) = (k-1)!\, du_1 \cdots du_{k-1}, \qquad b = (1, \ldots, 1). \tag{1}$$

If f is $k-1$ times continuously differentiable, $F^{(k-1)}(1, \ldots, 1; z_1, \ldots, z_k)$ is a divided difference when the z_i are all distinct and the derivative $f^{(k-1)}$ when the z_i are all equal. With this notation we shall unite Taylor's series and Newton's divided-difference interpolation formula. (Another way of uniting the two requires f to be holomorphic since contour integrals are used.)

Let Ω be either an open interval of the real line or a convex open set in the complex plane, and let the complex-valued function f be defined and continuous on Ω. If x and y are points in Ω, then by (5.1-1),

$$F(1, 1; x, y) = \int_0^1 f[ux + (1-u)y]\, du.$$

If $x \neq y$, we choose $t = ux + (1 - u)y$ as a new variable of integration:

$$F(1, 1; x, y) = \frac{1}{x - y} \int_y^x f(t)\, dt, \qquad x \neq y, \tag{2}$$

where the path of integration is the line segment joining x and y. If f is continuously differentiable on Ω in the real case or holomorphic in the complex case, we have similarly

$$F'(1, 1; x, y) = \frac{1}{x - y} \int_y^x f'(t)\, dt$$

and thus

$$F'(1, 1; x, y) = \frac{f(x) - f(y)}{x - y}, \qquad x \neq y, \tag{3}$$

Hence $F'(1, 1; x, y)$ is the first divided difference if $x \neq y$; if $x = y$ it is the derivative $f'(x)$ by (5.2-2). We now prove a generalization of (3).

LEMMA 5.5-1 Let $n \in \mathbb{N}$ and assume either

(i) Ω is an open interval in $\mathbb{R}$ and $f\colon \Omega \to \mathbb{C}$ is $n + 1$ times continuously differentiable, or

(ii) Ω is a convex open set in $\mathbb{C}$ and $f\colon \Omega \to \mathbb{C}$ is holomorphic.

Suppose $z_1, \ldots, z_n, x, y \in \Omega$ and consider $\{z_1, \ldots, z_n\}$ to be empty if $n = 0$. Then

$$\begin{aligned} &F^{(n)}(1, \ldots, 1; z_1, \ldots, z_n, x) - F^{(n)}(1, \ldots, 1; z_1, \ldots, z_n, y) \\ &\quad = \frac{x - y}{n + 1} F^{(n+1)}(1, \ldots, 1; z_1, \ldots, z_n, x, y). \end{aligned} \tag{4}$$

Proof We assume $n \geq 1$, since (3) is the case $n = 0$. By (5.3-1),

$$\begin{aligned} &F^{(n+1)}(1, \ldots, 1; z_1, \ldots, z_n, x, y) \\ &= (n + 1)! \int_{E_{n+1}} f^{(n+1)}[u \cdot z + u_{n+1}x + (1 - u_1 - \cdots - u_{n+1})y]\, du_1 \cdots du_{n+1}, \end{aligned}$$

where $u \cdot z = \sum_{i=1}^n u_i z_i$. Since (4) is trivial if $x = y$, we assume $x \neq y$ and replace u_{n+1} as a variable of integration by $t = u_{n+1}x + (1 - u_1 - \cdots - u_{n+1})y$. The limits of u_{n+1} are 0 and $1 - u_1 - \cdots - u_n$, and those of t are $\eta = (1 - u_1 - \cdots - u_n)y$ and $\xi = (1 - u_1 - \cdots - u_n)x$. Since the integrand is continuous and therefore bounded on the compact set E_{n+1}, the order of integration is immaterial by Fubini's theorem. Thus the right side is equal to

$$\begin{aligned}
&\frac{(n+1)!}{x-y}\int_{E_n}\int_\eta^\xi f^{(n+1)}(u\cdot z+t)\,dt\,du_1\cdots du_n \\
&\quad=\frac{(n+1)!}{x-y}\int_{E_n}[f^{(n)}(u\cdot z+\xi)-f^{(n)}(u\cdot z+\eta)]\,du_1\cdots du_n \\
&\quad=\frac{n+1}{x-y}[F^{(n)}(1,\ldots,1;z_1,\ldots,z_n,x)-F^{(n)}(1,\ldots,1;z_1,\ldots,z_n,y)]. \qquad \square
\end{aligned}$$

If $z_1, z_2, \ldots$ are distinct points in φ, the divided differences of f are defined by

$$\begin{aligned}
f[z_1,z_2]&=\frac{f(z_1)-f(z_2)}{z_1-z_2}, \\
f[z_1,\ldots,z_{n+1}]&=\frac{f[z_1,\ldots,z_n]-f[z_1,\ldots,z_{n-1},z_{n+1}]}{z_n-z_{n+1}}, \qquad n=2,3,\ldots.
\end{aligned} \tag{5}$$

Comparison with (3) and (4) shows that if $f\in C_p(\Omega)$,

$$f[z_1,\ldots,z_{n+1}]=\frac{1}{n!}F^{(n)}(1,\ldots,1;z_1,\ldots,z_{n+1}), \qquad n=1,\ldots,p. \tag{6}$$

The right side is symmetric in the z_i by Theorem 5.2-3. Also, (6) gives a meaning to the divided difference if the z_i are not all distinct; in particular it reduces to $f^{(n)}(a)/n!$ by (5.2-2) if $z_1=\cdots=z_{n+1}=a$.

THEOREM 5.5-2 (*Newton–Taylor series with remainder*) Assume that either

(i) Ω is an open interval of the real line and $f\colon\Omega\to\mathbb{C}$ is p times continuously differentiable, where $p\geq 1$, or

(ii) Ω is a convex open set in the complex plane and $f\colon\Omega\to\mathbb{C}$ is holomorphic.

In case (ii), p will denote any positive integer. If $x, z_1, \ldots, z_p\in\Omega$. then

$$\begin{aligned}
f(x)=f(z_1)&+\sum_{n=1}^{p-1}\frac{1}{n!}F^{(n)}(1,\ldots,1;z_1,\ldots,z_{n+1})(x-z_1)\cdots(x-z_n) \\
&+\frac{1}{p!}F^{(p)}(1,\ldots,1;z_1,\ldots,z_p,x)(x-z_1)\cdots(x-z_p).
\end{aligned} \tag{7}$$

Remark The general term of the sum is a polynomial of degree n in x, but in the remainder x occurs also as an argument of $F^{(p)}$. If the z_i are all distinct, the coefficient $F^{(n)}/n!$ is an nth divided difference; the expansion in this case was discovered by Newton. In the other extreme case, in which the z_i are all equal, say $z_1 = \cdots = z_p = a$, the expansion reduces by Theorems 5.2-3 and 5.2-4 to Taylor's series with remainder,

$$f(x) = \sum_{n=0}^{p-1} \frac{1}{n!} f^{(n)}(a)(x-a)^n + \frac{1}{p!} F^{(p)}(1, p; x, a)(x-a)^p. \tag{8}$$

Proof If $p = 1$, the summation is empty and (7) is equivalent to (3). If $p > 1$, we make the inductive hypothesis that (7) with p replaced by $p - 1$ is true. The remainder for this case is

$$\begin{aligned}
&\frac{1}{(p-1)!} F^{(p-1)}(1, \ldots, 1; z_1, \ldots, z_{p-1}, x)(x - z_1) \cdots (x - z_{p-1}) \\
&\quad = \frac{1}{(p-1)!} \bigg[F^{(p-1)}(1, \ldots, 1; z_1, \ldots, z_p) \\
&\qquad + \frac{x - z_p}{p} F^{(p)}(1, \ldots, 1; z_1, \ldots, z_p, x) \bigg] (x - z_1) \cdots (x - z_{p-1}),
\end{aligned}$$

where we have used (4). The first term in brackets gives the term in (7) with $n = p - 1$, and the second term gives the remainder in (7). It follows by induction that (7) is true for all values of p for which $f \in C_p(\Omega)$. □

We shall show now that the remainder in Taylor's series (8) is a p-fold repeated integral of $f^{(p)}$. We replace $f^{(p)}$ by f and assume that either

(i) $\Omega \subset \mathbb{R}$ is an open interval and $f: \Omega \to \mathbb{C}$ is continuous, or
(ii) $\Omega \subset \mathbb{C}$ is a convex open set and $f: \Omega \to \mathbb{C}$ is holomorphic.

If $a, x \in \Omega$, we define the function $I^n f: \Omega \to \mathbb{C}$ for every $n \in \mathbb{N}$ by

$$I^0 f(x) = f(x), \qquad I^n f(x) = \frac{(x-a)^n}{n!} F(1, n; x, a), \qquad n - 1 \in \mathbb{N}. \tag{9}$$

Then, for $n \geq 1$,

$$\begin{aligned}
I^n f(x) &= \frac{(x-a)^n}{n!} \int_0^1 f[ux + (1-u)a] \frac{(1-u)^{n-1}}{\mathrm{B}(1, n)} \, du \\
&= \int_a^x f(t) \frac{(x-t)^{n-1}}{(n-1)!} \, dt.
\end{aligned} \tag{10}$$

It follows that

$$I^n f(x) = \frac{d}{dx} I^{n+1} f(x), \qquad n \in \mathbb{N}. \tag{11}$$

Since $I^{n+1}f(a) = 0$, integration of (11) yields

$$I^{n+1}f(x) = \int_a^x I^n f(t)\,dt. \tag{12}$$

By iteration of (12),

$$I^n f(z_1) = \int_a^{z_1} dz_2 \int_a^{z_2} dz_3 \cdots \int_a^{z_{n-1}} dz_n \int_a^{z_n} f(t)\,dt. \tag{13}$$

The function $I^n f$ is an n-fold repeated integral of f, and we may think of I as an integration operator.

One can extend (9) to nonintegral powers of I by defining

$$I^\nu f(x) = (x-a)^\nu \frac{F(1, \nu; x, a)}{\Gamma(1+\nu)}, \tag{14}$$

where $x > a$ in case (i) and $|\arg(x-a)| < \pi$ in case (ii). The representation corresponding to (10) is called a Riemann–Liouville fractional integral of order ν:

$$I^\nu f(x) = \frac{1}{\Gamma(\nu)} \int_a^x f(t)(x-t)^{\nu-1}\,dt, \qquad \nu \in \mathbb{C}_>. \tag{15}$$

Although the integral has a meaning only if $\nu \in \mathbb{C}_>$, we shall see in Section 6.3 and Exercise 6.3-3 that $I^\nu f$ can be continued analytically throughout the ν plane [in case (ii); for case (i) see M. Riesz (1949)] and assumes the values

$$I^{-n}f(x) = f^{(n)}(x), \qquad n \in \mathbb{N}. \tag{16}$$

Thus I^{-n} is a differentiation operator, and $I^\nu f$ may be called either a fractional integral of order ν or a fractional derivative of order $-\nu$.

5.6 Associated functions

Define the unit vectors

$$e_1 = (1, 0, \ldots, 0), e_2 = (0, 1, 0, \ldots, 0), \ldots, e_k = (0, \ldots, 0, 1), \tag{1}$$

and denote a k-tuple of integers by

$$m = (m_1, \ldots, m_k) = \sum_{i=1}^k m_i e_i.$$

Thus $b - e_i = (b_1, \ldots, b_i - 1, \ldots, b_k)$, and $b + m = (b_1 + m_1, \ldots, b_k + m_k)$. If $n \in \mathbb{N}$ and $m \in \mathbb{Z}^k$, the functions $F(b, z)$ and $F^{(n)}(b+m, z)$ are called associated functions. Several functions are said to be associated if they are pairwise associated. For instance $F(b, z)$, $F'(b + e_i, z)$, and $F''(b - e_j, z)$ are associated functions. Algebraic and differential relations between associated functions are very useful, for example in deriving recurrence relations between Legendre functions or reducing elliptic integrals to standard forms.

If $b \in \mathbb{C}_>{}^k$, define c and $w = (w_1, \ldots, w_k)$ by

$$c = \sum_{i=1}^{k} b_i, \qquad w = b/c = (b_1/c, \ldots, b_k/c). \tag{2}$$

Since the b_i have positive real parts, c also has a positive real part. The components $w_i = b_i/c$ have unit sum and may be regarded as weights if the b_i are real and positive. It follows from (4.4-8) that

$$w_i = \int_E u_i \, d\mu_b(u), \qquad i = 1, \ldots, k. \tag{3}$$

RELATIONS 5.6-1 Let f satisfy the assumptions of either Theorem 5.3-2 or 5.3-3, let $z \in \Omega^k$ and $b \in \mathbb{C}_>{}^k$, and define c and w by (2). Then

$$F(b, z) = \sum_{i=1}^{k} w_i F(b + e_i, z). \tag{4}$$

If f' exists and is continuous, then

$$D_i F(b, z) = w_i F'(b + e_i, z). \tag{5}$$

More generally, if $f^{(n)}$ exists and is continuous and if $m_1, \ldots, m_k$ are nonnegative integers whose sum is n, then

$$D_1^{m_1} \cdots D_k^{m_k} F(b, z) = \frac{(b_1, m_1) \cdots (b_k, m_k)}{(c, n)} F^{(n)}(b + m, z). \tag{6}$$

Proof By (4.4-8),

$$u_1^{m_1} \cdots u_k^{m_k} \, d\mu_b(u) = \frac{\mathrm{B}(b + m)}{\mathrm{B}(b)} \, d\mu_{b+m}(u) = \frac{(b_1, m_1) \cdots (b_k, m_k)}{(c, n)} \, d\mu_{b+m}(u). \tag{7}$$

In particular,

$$u_i \, d\mu_b(u) = w_i \, d\mu_{b+e_i}(u). \tag{8}$$

Since $\sum_{i=1}^{k} u_i = 1$, (8) implies

$$d\mu_b(u) = \sum_{i=1}^{k} w_i \, d\mu_{b+e_i}(u), \tag{9}$$

from which (4) follows at once by (5.2-1). Equations (7) and (5.3-2) imply (6), and (5) is a special case of (6). □

RELATION 5.6-2 Let f satisfy the assumptions of either Theorem 5.3-2, with $n \geq 1$, or Theorem 5.3-3. Let $z \in \Omega^k$, $b - e_i - e_j \in \mathbb{C}_>{}^k$ for some i and j, and $c = \sum_{i=1}^k b_i$. Then

$$(z_i - z_j)F'(b, z) + (c - 1)[F(b - e_i, z) - F(b - e_j, z)] = 0. \tag{10}$$

Proof Let a be a fixed point in Ω and define, for every $x \in \Omega$,

$$g(x) = \int_a^x f(t)\,dt.$$

Then $g' = f$ and $g'' = f'$ exist and are continuous on Ω. Assume $b \in \mathbb{C}_>{}^k$ and let $G(b, z)$ denote the Dirichlet average of g. By Theorem 5.4-1,

$$(z_i - z_j)D_i D_j G + b_i D_j G - b_j D_i G = 0,$$

and by (6) this implies

$$(z_i - z_j)\frac{b_i b_j}{c(c+1)} G''(b + e_i + e_j, z) + \frac{b_i b_j}{c} G'(b + e_j, z) - \frac{b_i b_j}{c} G'(b + e_i, z) = 0.$$

Since $G' = F$ and $G'' = F'$, replacement of b by $b - e_i - e_j$ (which entails replacement of c by $c - 2$) yields (10). □

RELATION 5.6-3 Let f satisfy the assumptions of either Theorem 5.3-2, with $n \geq 1$, or Theorem 5.3-3. Let $z \in \Omega^k$, $k \geq 3$, and $b - e_h - e_i - e_j \in \mathbb{C}_>{}^k$ for some h, i, j. Then

$$(z_i - z_j)F(b - e_h, z) + (z_j - z_h)F(b - e_i, z) + (z_h - z_i)F(b - e_j, z) = 0. \tag{11}$$

Proof Write (10) with indices i and j, then with indices j and h, and finally with indices h and i. Multiply the three equations by z_h, z_i, and z_j, respectively, and add. □

Equation (10) states that the quantity

$$z_i F'(b, z) + (c - 1)F(b - e_i, z) \tag{12}$$

is independent of i for all values of i such that $\operatorname{Re} b_i > 1$. The following relation between associated functions gives an expression for this quantity which is manifestly independent of i.

RELATION 5.6-4 Let f satisfy the assumptions of either Theorem 5.3-2, with $n \geq 1$, or Theorem 5.3-3. Let $z \in \Omega^k$, $b - e_i \in \mathbb{C}_>{}^k$ or some value of i, and $c = \sum_{i=1}^k b_i$. Then

$$z_i F'(b, z) + (c - 1)F(b - e_i, z) = \sum_{j=1}^k z_j D_j F(b, z) + (c - 1)F(b, z). \tag{13}$$

Proof In (10) replace b by $b + e_j$, which entails replacing c by $c + 1$. Then

$$(z_i - z_j)F'(b + e_j, z) + c[F(b - e_i + e_j, z) - F(b, z)] = 0,$$

where $b - e_i \in \mathbb{C}_>{}^k$. Multiply this equation by $w_j = b_j/c$ and sum with respect to j from 1 to k. We apply (5) and find

$$\sum_{j=1}^{k} (z_i - z_j)D_j F(b, z) + \sum_{j=1}^{k} (b - e_i)_j[F(b - e_i + e_j, z) - F(b, z)] = 0,$$

because $(b - e_i)_j = b_j$ if $j \neq i$ and the quantity in brackets vanishes if $j = i$. Carrying out the second summation by (4), we obtain

$$\sum_{j=1}^{k} (z_i - z_j)D_j F(b, z) + (c - 1)F(b - e_i, z) - (c - 1)F(b, z) = 0. \quad (14)$$

According to (5.3-3) this is equivalent to (13). □

Remark In Section 6.3 we shall extend the definition of $F(b, z)$ to values of the b_i which do not have positive real parts. It will then be possible to relax the conditions imposed in Relations 5.6-1 and 5.6-4. For example, the requirement that $b - e_i \in \mathbb{C}_>{}^k$ can be replaced by $c - 1 \in U$.

5.7 Averages of power series

A function f which is analytic on an open circular disk in the complex plane can be represented by a power series. The Dirichlet average of f has a series representation with the same coefficients, in which powers are replaced by homogeneous polynomials. This representation will be used in Sections 5.8 and 5.9 to connect Dirichlet averages of e^x and x^t with hypergeometric series. Later it will provide an analytic continuation of $F(b, z)$ to values of the parameters with negative real parts.

DEFINITION 5.7-1 Let $n \in \mathbb{N}$ and let μ_b be a Dirichlet measure on the standard simplex E in $\mathbb{R}^{k-1}$, $k \geq 2$. For every $z \in \mathbb{C}^k$ define

$$R_n(b, z) = \int_E (u \cdot z)^n \, d\mu_b(u). \quad (1)$$

If $k = 1$, define $R_n(b, z) = z^n$.

This Dirichlet average of x^n will be discussed in detail in Chapter 6, and for the moment a few remarks will suffice. Note that

$$R_0(b, z) = 1 \qquad \text{and} \qquad R_1(b, z) = \sum_{i=1}^{k} w_i z_i. \quad (2)$$

Since $(u \cdot z)^n$ is a homogeneous polynomial of degree n in $z_1, \ldots, z_k$, so is $R_n(b, z)$. According to (5.2-2) the monomial x^n is the restriction of R_n to the diagonal of $\mathbb{C}^k$:

$$R_n(b_1, \ldots, b_k; x, \ldots, x) = x^n. \tag{3}$$

REPRESENTATION 5.7-2 Let $\Omega \subset \mathbb{C}$ be an open disk with center λ, and let $f\colon \Omega \to \mathbb{C}$ be holomorphic. If $(b, z) \in \mathbb{C}_>{}^k \times \Omega^k$, then the Dirichlet average of f has the series representation

$$F(b, z) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(\lambda) R_n(b, z - \lambda), \tag{4}$$

where $z - \lambda = (z_1 - \lambda, \ldots, z_k - \lambda)$.

Remark The restriction of this series to the diagonal of Ω^k is the Taylor series of f.

Proof Since f is holomorphic, it can be represented on Ω by its Taylor series,

$$f(x) = \sum_{n=0}^{\infty} a_n (x - \lambda)^n, \qquad a_n = \frac{1}{n!} f^{(n)}(\lambda). \tag{5}$$

The series converges uniformly on any compact subset of Ω. For any fixed $z \in \Omega^k$, let $u \cdot z$ be a convex combination and define $u' = (u_1, \ldots, u_{k-1})$. Then $u' \mapsto u \cdot z$ is a mapping of the set E onto the compact set $\operatorname{con}(z) \subset \Omega$ (see Section 5.1). Since (5) converges uniformly on $\operatorname{con}(z)$, the series

$$f(u \cdot z) = \sum_{n=0}^{\infty} a_n (u \cdot z - \lambda)^n = \sum_{n=0}^{\infty} a_n [u \cdot (z - \lambda)]^n$$

converges uniformly on E. Term-by-term integration over E with respect to the measure μ_b yields (4). □

The following special case of (4) will be useful in the next two sections.

REPRESENTATION 5.7-3 Let $\Omega \subset \mathbb{C}$ be an open disk with center λ, and let $f\colon \Omega \to \mathbb{C}$ be holomorphic. If $x \in \Omega$ and $\operatorname{Re} \gamma > \operatorname{Re} \beta > 0$, then

$$F(\beta, \gamma - \beta; x, \lambda) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(\lambda) \frac{(\beta, n)}{(\gamma, n)} (x - \lambda)^n. \tag{6}$$

Proof By (4),

$$F(\beta, \gamma - \beta; x, \lambda) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(\lambda) R_n(\beta, \gamma - \beta; x - \lambda, 0). \tag{7}$$

Equations (2)–(4) are exact analogs of properties of the exponential function. In particular we note the differential equations [see (2)]

$$\frac{d}{dx}(e^x) = e^x \qquad \text{and} \qquad \sum_{i=1}^{k} \frac{\partial S}{\partial z_i} = S. \tag{5}$$

The first is equivalent to the relation $e^{x+\lambda} = e^\lambda e^x$ and the second to (3) [see Exercise 5.8-1].

We shall now show that in the case $k = 2$ the S function is equivalent to the confluent hypergeometric function ${}_1F_1$ defined by (2.4-17).

FORMULA 5.8-5 If $x, y \in \mathbb{C}$ and $\operatorname{Re} \gamma > \operatorname{Re} \beta > 0$, then

$$S(\beta, \gamma - \beta; x, y) = e^y \, {}_1F_1(\beta; \gamma; x - y). \tag{6}$$

Proof Equation (5.7-6) with $f(x) = e^x$ and $\lambda = 0$ is

$$S(\beta, \gamma - \beta; x, 0) = \sum_{n=0}^{\infty} \frac{(\beta, n)}{(\gamma, n)} \frac{x^n}{n!} = {}_1F_1(\beta; \gamma; x). \tag{7}$$

Replace x by $x - y$, multiply by e^y, and use (3). □

Equation (7) expresses a ${}_1F_1$ series as a Dirichlet average of e^x over a line segment with one endpoint at the origin. The symmetry property of S (Theorem 5.2-3) implies that each side of (6) is invariant under interchange of x with y and β with $\gamma - \beta$. Thus

$$e^y \, {}_1F_1(\beta; \gamma; x - y) = e^x \, {}_1F_1(\gamma - \beta; \gamma; y - x). \tag{8}$$

Setting $y = 0$, we have

$${}_1F_1(\beta; \gamma; x) = e^x \, {}_1F_1(\gamma - \beta; \gamma; -x). \tag{9}$$

Equation (9) is an important property of ${}_1F_1$ known as Kummer's transformation or Kummer's first formula. Our proof shows it to be an expression of permutation symmetry, hidden or at least diguised by conventional notation. In the next section and in later chapters we shall find other transformation formulas which express hidden symmetries. Formulas of this kind are eliminated by using explicitly symmetric functions such as the S function.

The function ${}_1F_1$ satisfies a linear differential equation of second order known as the confluent hypergeometric equation. It is interesting to see how this is related to the Euler–Poisson equation satisfied by $S = S(\beta, \beta'; x, y)$. Because of (5) we have two partial differential equations for S,

$$(x - y)\frac{\partial^2 S}{\partial x \, \partial y} + \beta \frac{\partial S}{\partial y} - \beta' \frac{\partial S}{\partial x} = 0 \qquad \text{and} \qquad \frac{\partial S}{\partial x} + \frac{\partial S}{\partial y} = S. \tag{10}$$

We eliminate $\partial S/\partial y$ between these two equations and find

$$(x-y)\frac{\partial^2 S}{\partial x^2} + (\beta + \beta' - x + y)\frac{\partial S}{\partial x} - \beta S = 0. \tag{11}$$

With (7) in mind, we set $\beta' = \gamma - \beta$ and $y = 0$. Then the function

$$v(x) = S(\beta, \gamma - \beta; x, 0) = {}_1F_1(\beta; \gamma; x)$$

satisfies

$$xv'' + (\gamma - x)v' - \beta v = 0. \tag{12}$$

The confluent hypergeometric equation (12) occurs frequently in applied mathematics because any differential equation of the form

$$(a_0 x + b_0)y'' + (a_1 x + b_1)y' + (a_2 x + b_2)y = 0, \tag{13}$$

where the a_i and b_i are constants, can be transformed into (12) by suitable change of the independent and dependent variables (Erdélyi, 1953, Vol. 1, p. 249). The same is true of the equation

$$y'' + \left(a + \frac{b}{x}\right)y' + \left(\alpha + \frac{\beta}{x} + \frac{\gamma}{x^2}\right)y = 0. \tag{14}$$

A special case of (14) is Bessel's equation,

$$x^2 y'' + xy' + (x^2 - v^2)y = 0, \tag{15}$$

which results from separating Laplace's equation in cylindrical coordinates. A related example is the radial Schrödinger equation for a particle in a Coulomb field.

In many cases solutions can be found directly from the results in Section 5.4. In (5.4-11), for example, we choose $f(\zeta) = e^{ik\zeta}$, where k is a constant, and use Theorem 5.2-6, which implies that the Dirichlet average is $S(b, ikz)$. Thus a particular solution of the wave equation is

$$W = S\left[\frac{n-1}{2}, \frac{n-1}{2}; ik(t+r), ik(t-r)\right].$$

Using (3) yields $W = e^{ikt}\psi$, where ψ is independent of t. Therefore, (5.4-11) and (5.4-12) become, for $n \geq 2$,

$$\psi = S\left(\frac{n-1}{2}, \frac{n-1}{2}; ikr, -ikr\right), \qquad r = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}, \tag{16}$$

$$\sum_{i=1}^{n} \frac{\partial^2 \psi}{\partial x_i^2} + k^2\psi = 0. \tag{17}$$

Equation (17) is Helmholtz's equation. We now replace n by $n+1$ and set $x_{n+1} = it$. Thus, for $n \geq 1$,

$$\chi = S(n/2, n/2; iks, -iks), \qquad s = \left(\sum_{i=1}^{n} x_i^2 - t^2\right)^{1/2} = (r^2 - t^2)^{1/2}, \tag{18}$$

$$\sum_{i=1}^{n} \frac{\partial^2 \chi}{\partial x_i^2} + k^2 \chi = \frac{\partial^2 \chi}{\partial t^2}. \tag{19}$$

Equation (19) is the Klein–Gordon equation.

If we set $f(\zeta) = e^{ik\zeta}$ and $\varphi = e^{ikx}\omega$ in (5.4-19) and (5.4-20), we find

$$\omega = y^{-\lambda \pm \mu} S(\tfrac{1}{2} \pm \mu + \nu, \tfrac{1}{2} \pm \mu - \nu; iky, -iky), \tag{20}$$

$$\omega'' + \frac{2\lambda + 1}{y}\omega' + \left(k^2 - \frac{2ik\nu}{y} + \frac{\lambda^2 - \mu^2}{y^2}\right)\omega = 0. \tag{21}$$

The case $\lambda = \frac{1}{2}$ is the radial Schrödinger equation of a particle in a Coulomb field. The quantity $ik\nu$ is real in this case and measures the strength of the field. In an unbound state the wave number k is real and so is the Born parameter $\eta = i\nu$. Setting $\mu = L + \frac{1}{2}$, where the nonnegative integer L is the quantum number of angular momentum, we find that the regular Coulomb wave function of a particle with radial coordinate y is

$$y^L S(1 + L - i\eta, 1 + L + i\eta; iky, -iky). \tag{22}$$

Since the exponent difference 2μ is an integer in this case, choosing the lower signs in (20) does not give the second solution (cf. Section 5.12).

If $\lambda = \nu = 0$ and $k = 1$, (21) becomes Bessel's equation [see (15)], and the Bessel function denoted by J_μ is the solution

$$J_\mu(y) = \frac{(y/2)^\mu}{\Gamma(1 + \mu)} S(\tfrac{1}{2} + \mu, \tfrac{1}{2} + \mu; iy, -iy). \tag{23}$$

Comparison of (23) with (22) shows that the radial Schrödinger equation of a free particle ($\eta = 0$) has the solution $j_L(ky)$, where

$$j_L(x) = (\pi/2x)^{1/2} J_{L+1/2}(x) = \frac{(x/2)^L}{(\frac{3}{2}, L)} S(1 + L, 1 + L; ix, -ix), \qquad L \in \mathbb{N}. \tag{24}$$

A Bessel function of order equal to half an odd integer is called a spherical Bessel function because it occurs in problems with spherical rather than cylindrical symmetry.

Because Bessel's equation has a singular point at the origin, the Bessel function J_μ has in general a branch point at the origin. The S function on the right side of (23) is J_μ with its branch point removed. The S function is entire in y and is normalized to 1 at the origin, like J_0.

5.9 Averages of x^t

Unless t is a nonnegative integer, x^t has either a branch point or a pole at $x = 0$. In averaging x^t over the convex hull of several points in the plane, one must require that the convex hull does not contain the origin if the integral is to be well defined. The exceptional cases $t = 0, 1, 2, \ldots$ have already been introduced in Definition 5.7-1.

DEFINITION 5.9-1 Let μ_b be a Dirichlet measure on the standard simplex $E \subset \mathbb{R}^{k-1}$, $k \geq 2$. Let H be a half-plane in $\mathbb{C} - \{0\}$. Let $\Omega = H$ if $t \in \mathbb{C} - \mathbb{N}$, but if $t \in \mathbb{N}$, let $\Omega = \mathbb{C}$. For every $z \in \Omega^k$ define

$$R_t(b, z) = \int_E (u \cdot z)^t \, d\mu_b(u). \tag{1}$$

An alternative notation is $R(-t, b, z) = R_t(b, z)$. If $k = 1$, define $R_t(b, z) = z^t$.

Remark If $\zeta = u \cdot z \in H$, a single-valued branch of $\arg \zeta$ determines a single-valued branch of ζ^t. We assume $\arg \zeta$ to be continuous on H but do not specify a particular branch at this time. Different choices yield different branches of R_t.

THEOREM 5.9-2 Let H be an open half-plane in $\mathbb{C} - \{0\}$. Then $R_t(b, z)$ is holomorphic in (t, b, z) on $\mathbb{C} \times \mathbb{C}_>{}^k \times H^k$ and can be differentiated under the integral sign in (1). If $D_i = \partial/\partial z_i$, then

$$\sum_{i=1}^{k} D_i R_t = t R_{t-1} \qquad \text{and} \qquad \sum_{i=1}^{k} z_i D_i R_t = t R_t. \tag{2}$$

Proof By Theorem 5.3-3, $R_t(b, z)$ is holomorphic in (b, z) on $\mathbb{C}_>{}^k \times H^k$ for every $t \in \mathbb{C}$, and Eqs. (2) are special cases of (5.3-3) and (5.3-4). Since a function is holomorphic in several variables jointly if it is holomorphic in each one separately, we need only prove that $R_t(b, z)$ is entire in t for each fixed $(b, z) \in \mathbb{C}_>{}^k \times H^k$. In Theorem B.4 we choose $\mu = \mu_b$, $T = E$, $Z = \mathbb{C}$, and verify the three hypotheses of that theorem. In the following, $u' = (u_1, \ldots, u_{k-1}) \in E$ and t denotes the index of R_t rather than the variable of integration used in Theorem B.4.

(i) $(u \cdot z)^t$ is entire in t for every $u' \in E$ because $u \cdot z \in H$ implies $u \cdot z \neq 0$.

(ii) $u' \mapsto (u \cdot z)^t$ is continuous and bounded on the compact set E for every fixed $t \in \mathbb{C}$ (and every fixed $z \in H^k$). A continuous bounded function is in $L_1(|\mu_b|)$ by (4.4-4).

(iii) If K is a compact set in $\mathbb{C}$, the function $(t, u') \mapsto (u \cdot z)^t$ is continuous on the compact set $K \times E$ and is therefore bounded. We can choose g_K to be a constant.

It follows from Theorem B.4 that $R_t(b, z)$ is entire in t for each fixed $(b, z) \in \mathbb{C}_>{}^k \times H^k$ and can be differentiated under the integral sign in (1). ▯

FORMULA 5.9-3 Suppose $\lambda \in \mathbb{C} - \{0\}$. Define $R_t(b, z)$ according to Definition 5.9-1 for every $z \in \Omega^k$, let $\lambda z = (\lambda z_1, \ldots, \lambda z_k)$, and define $R_t(b, \zeta)$ for every $\zeta \in (\Omega')^k$, where $\Omega' = \{\lambda z : z \in \Omega\}$. Then

$$R_t(b, \lambda z) = \lambda^t R_t(b, z). \tag{3}$$

Remark Equation (3) follows from (1). If we choose arg λ and a single-valued branch of $R_t(b, z)$, (3) holds for a unique single-valued branch of $R_t(b, \lambda z)$. The situation is the same as for the analogous equation $(\lambda x)^t = \lambda^t x^t$.

REPRESENTATION 5.9-4 Let $a \in \mathbb{C}$ and $b \in \mathbb{C}_>{}^k$. If Ω denotes the open unit disk in $\mathbb{C}$, let $z \in \Omega^k$ and $1 - z = (1 - z_1, \ldots, 1 - z_k)$. Then

$$R_{-a}(b, 1 - z) = \sum_{n=0}^{\infty} \frac{(a, n)}{n!} R_n(b, z). \tag{4}$$

Proof In Representation 5.7-2 let Ω be the open unit disk, $\lambda = 0$, and $f(x) = (1 - x)^{-a}$. Then $F(b, z) = R_{-a}(b, 1 - z)$ by Theorem 5.2-6. ▯

Equations (2)–(4) are exact analogs of the following properties of x^t:

$$\frac{d}{dx}(x^t) = tx^{t-1} \qquad \text{and} \qquad x\frac{d}{dx}(x^t) = tx^t,$$

$$(\lambda x)^t = \lambda^t x^t, \qquad (1 - x)^{-a} = \sum_{n=0}^{\infty} \frac{(a, n)}{n!} x^n, \qquad x \in \Omega.$$

The last of these is the binomial series (2.4-11). The homogeneity relation (3) is equivalent to the second equation of (2), which is Euler's differential equation for a homogeneous function of degree t.

The R function has the properties deduced earlier in this chapter for the Dirichlet average of any analytic function, in particular the symmetry property of Theorem 5.2-3 and the Euler–Poisson differential equations.

Equation (5.6-13) is greatly simplified by Euler's differential equation and becomes a relation between only three associated R functions.

RELATIONS 5.9-5 Let the assumptions of Definition 5.9-1 be satisfied, and let c and w be defined by (5.6-2). Then the following relations hold between associated R functions:

$$R_t(b, z) = \sum_{i=1}^{k} w_i R_t(b + e_i, z), \tag{5}$$

$$R_{t+1}(b, z) = \sum_{i=1}^{k} w_i z_i R_t(b + e_i, z), \tag{6}$$

$$cR_t(b, z) = (c + t)R_t(b + e_i, z) - tz_i R_{t-1}(b + e_i, z), \qquad i = 1, \ldots, k. \tag{7}$$

Proof Equation (5) is a special case of (5.6-4). To prove (6) we write

$$R_{t+1}(b, z) = \sum_{i=1}^{k} z_i \int_E u_i(u \cdot z)^t \, d\mu_b(u),$$

and use (5.6-8). To prove (7) we deduce from (5.6-13) and (2) that

$$(c - 1)R_t(b - e_i, z) = (c + t - 1)R_t(b, z) - tz_i R_{t-1}(b, z) \tag{8}$$

for every value of i satisfying $\operatorname{Re}(b - e_i) > 0$. Replacement of b by $b + e_i$ yields (7). □

RELATIONS 5.9-6 Let the assumptions of Definition 5.9-1 be satisfied, and let c and w be defined by (5.6-2). Then, for $i = 1, \ldots, k$, with $D_i = \partial/\partial z_i$,

$$D_i R_t(b, z) = w_i t R_{t-1}(b + e_i, z), \tag{9}$$

$$(z_i D_i + b_i) R_t(b, z) = w_i(c + t) R_t(b + e_i, z). \tag{10}$$

Proof Equation (9) is a special case of (5.6-5). Multiply (9) by z_i and (7) by w_i, and add to prove (10). □

In the remainder of this section we shall establish several relations which are peculiar to the important case $k = 2$. First, the R function is equivalent to Gauss's hypergeometric function, which is defined by (2.4-2) on the open unit disk and continued analytically by Exercise 4.4-2.

PROPOSITION 5.9-7 Let $\alpha, \beta, \gamma, x, y \in \mathbb{C}$ and assume $\operatorname{Re} \gamma > \operatorname{Re} \beta > 0$. Let H be a half-plane in $\mathbb{C} - \{0\}$. Assume $x, y \in \mathrm{H}$ if $\alpha \in U$, but if $-\alpha \in \mathbb{N}$ assume $y \neq 0$. Then

$$R_{-\alpha}(\beta, \gamma - \beta; x, y) = y^{-\alpha} \, {}_2F_1(\alpha, \beta; \gamma; 1 - x/y). \tag{11}$$

Proof In Representation 5.7-3 let Ω be the open unit disk, $\lambda = 0$, and $f(\zeta) = (1-\zeta)^{-\alpha}$. Then $F(b, z) = R_{-\alpha}(b, 1-z)$, and (5.7-6) becomes

$$R_{-\alpha}(\beta, \gamma-\beta; 1-\zeta, 1) = \sum_{n=0}^{\infty} \frac{(\alpha, n)(\beta, n)}{(\gamma, n)n!} \zeta^n = {}_2F_1(\alpha, \beta; \gamma; \zeta), \tag{12}$$

where $|\zeta| < 1$. If x and y satisfy $|x-y| < |y|$ as well as the assumptions stated above, we may set $\zeta = 1 - x/y$. We then multiply (12) by $y^{-\alpha}$ and use (3) to obtain (11). The additional restriction on x and y can be removed by the permanence of functional relations [see Ahlfors (1966, p. 277)], which assures that a functional relation between holomorphic functions holds also between their holomorphic continuations. □

By the permutation symmetry of R (Theorem 5.2-3), (11) implies

$$y^{-\alpha}\, {}_2F_1(\alpha, \beta; \gamma; 1-x/y) = x^{-\alpha}\, {}_2F_1(\alpha, \gamma-\beta; \gamma; 1-y/x) \tag{13}$$

or equivalently

$${}_2F_1(\alpha, \beta; \gamma; \zeta) = (1-\zeta)^{-\alpha}\, {}_2F_1\left(\alpha, \gamma-\beta; \gamma; \frac{\zeta}{\zeta-1}\right), \tag{14}$$

where $|\mathrm{ph}(1-\zeta)| < \pi$. The transformation (14), due to Pfaff, is therefore an expression of permutation symmetry.

The function ${}_2F_1$ satisfies a linear differential equation of second order known as the hypergeometric equation. It can be extracted from the Euler–Poisson equation in the same way as the confluent hypergeometric equation. In place of (5.8-10) the two partial differential equations satisfied by $R = R_{-\alpha}(\beta, \beta'; x, y)$ are

$$(x-y)\frac{\partial^2 R}{\partial x\, \partial y} + \beta \frac{\partial R}{\partial y} - \beta' \frac{\partial R}{\partial x} = 0 \qquad \text{and} \qquad x\frac{\partial R}{\partial x} + y\frac{\partial R}{\partial y} = -\alpha R. \tag{15}$$

Eliminating $\partial R/\partial y$ between them yields

$$x(x-y)\frac{\partial^2 R}{\partial x^2} + [\beta x + \beta' y + (\alpha+1)(x-y)]\frac{\partial R}{\partial x} + \alpha\beta R = 0. \tag{16}$$

In view of (12) we set $\beta' = \gamma - \beta$, $y = 1$, and $x = 1 - z$. Then the function

$$v(z) = R_{-\alpha}(\beta, \gamma-\beta; 1-z, 1) = {}_2F_1(\alpha, \beta; \gamma; z)$$

satisfies

$$z(1-z)v'' + [\gamma - (\alpha+\beta+1)z]v' - \alpha\beta v = 0. \tag{17}$$

The most important case of this hypergeometric equation is Legendre's differential equation,

$$(1-x^2)y'' - 2xy' + \left[\nu(\nu+1) - \frac{\mu^2}{1-x^2}\right]y = 0, \tag{18}$$

which is obtained by substituting $\alpha = \mu - \nu$, $\beta = \mu + \nu + 1$, $\gamma = \mu + 1$, $z = (1 - x)/2$, and $v = (x^2 - 1)^{-\mu/2}y$. The solutions of (18) are called Legendre functions. They occur throughout applied mathematics, often with $\theta = \arccos x$ as independent variable, because separation of Laplace's equation in spherical polar coordinates (r, θ, φ) leads to (18).

PROPOSITION 5.9-8 Let $\alpha, \alpha' \in \mathbb{C}$ and $\beta, \beta' \in \mathbb{C}_>$, and assume $\alpha + \alpha' = \beta + \beta'$. Let H be a half-plane in $\mathbb{C} - \{0\}$ and assume $x, y \in H$. Then

$$R_{-\alpha}(\beta, \beta'; x, y) = y^{\beta-\alpha} R_{-\beta}(\alpha, \alpha'; x, y), \qquad \alpha, \alpha' \in \mathbb{C}_>, \tag{19}$$

$$R_{-\alpha}(\beta, \beta'; x, y) = x^{\beta'-\alpha} R_{-\beta'}(\alpha', \alpha; x, y), \qquad \alpha, \alpha' \in \mathbb{C}_>, \tag{20}$$

$$R_{-\alpha}(\beta, \beta'; x, y) = x^{\beta'-\alpha} y^{\beta-\alpha} R_{-\alpha'}(\beta', \beta; x, y), \tag{21}$$

$$R_{-\beta-\beta'}(\beta, \beta'; x, y) = x^{-\beta} y^{-\beta'}. \tag{22}$$

Remark The first three transformations, which are peculiar to the case $k = 2$, allow any one of four parameters to be the degree of homogeneity of the R function without changing the arguments x and y. Requirements that $\alpha, \alpha' \in \mathbb{C}_>$ and $\beta, \beta' \in \mathbb{C}_>$ will be relaxed in Section 6.3 to the condition $\alpha + \alpha' = \beta + \beta' \in U$.

Proof If $x, y \in H$ and $|x - y| < |y|$, Proposition 5.9-7 implies

$$y^{\alpha} R_{-\alpha}(\beta, \beta'; x, y) = \sum_{n=0}^{\infty} \frac{(\alpha, n)(\beta, n)}{(\beta + \beta', n) n!} \left(1 - \frac{x}{y}\right)^n. \tag{23}$$

Since the series is unaltered by interchanging α with β and α' with β', (19) is true provided we require $\alpha, \alpha' \in \mathbb{C}_>$ so that the right side of (19) has a meaning. The restriction $|x - y| < |y|$ can be removed by the permanence of functional relations. Now interchange β with β' and x with y in (19). By permutation symmetry (Theorem 5.2-3) the left side is unchanged while the right side becomes $x^{\beta'-\alpha} R_{-\beta'}(\alpha, \alpha'; y, x)$, which is equal to the right side of (20) by permutation symmetry. Combining (19) with (20) yields (21) if $\alpha, \alpha' \in \mathbb{C}_>$. Since α and α' do not occur as b parameters in (21), this condition can be removed by Theorem 5.9-2 and the permanence of functional relations. Set $\alpha' = 0$ in (21) to prove (22). ▯

RELATIONS 5.9-9 Let the assumptions of Definition 5.9-1 be satisfied; let $k = 2$, and let c and w be defined by (5.6-2). Then

$$w_i(z_j - z_i) R_t(b + e_i, z) = z_j R_t(b, z) - R_{t+1}(b, z), \qquad i = 1, 2; \quad j = 3 - i, \tag{24}$$

$$(c + t) R_{t+1}(b, z) - \sum_{i=1}^{2} (b_i + t) z_i R_t(b, z) + t z_1 z_2 R_{t-1}(b, z) = 0. \tag{25}$$

Proof Equations (5) and (6) may be solved for two unknowns, $R_t(b+e_1, z)$ and $R_t(b+e_2, z)$, to obtain (24). Substitution of (24) in (7) yields (25). ▯

5.10 Confluence

The exponential function and the power function are related by the limit formula

$$e^x = \lim_{t\to\infty}\left(1+\frac{x}{t}\right)^t, \tag{1}$$

where $x\in\mathbb{C}$ and t tends to infinity in any manner in $\mathbb{C}$. This formula is the simplest example of a process which is called confluence because it makes two singular points flow together. The branch points of $x\mapsto(1+x/t)^t$ are $-t$ and the point at infinity. (There is only one point at infinity for the complex plane, corresponding in stereographic projection to the north pole of the Riemann sphere.) As $t\to\infty$ the two branch points coalesce to form the essential singularity of the exponential function at infinity.

THEOREM 5.10-1 Let $t\in\mathbb{C}$, $b\in\mathbb{C}_>{}^k$, and $z\in\mathbb{C}^k$. Then

$$S(b, z) = \lim_{t\to\infty} R_t(b, 1+z/t), \tag{2}$$

where $1+z/t = (1+z_1/t, \ldots, 1+z_k/t)$.

Remark Equation (2) is an exact analog of (1), which is the case $z_1 = \cdots = z_k = x$. The S function may be called a confluent limit or confluent form of the R function.

Proof We first prove (1) for complex x and t. Given any fixed $x\in\mathbb{C}$, choose a positive number $r\geq 2|x|$, and define $A=\{t\in\mathbb{C}: |t|\geq r\}$. For every $t\in A$ the binomial series (2.2-4) is

$$\left(1+\frac{x}{t}\right)^t = \sum_{n=0}^{\infty}\frac{(-t,n)}{n!}\left(\frac{-x}{t}\right)^n. \tag{3}$$

By Exercise 2.2-10, $|t|\geq r$ implies

$$\left|\frac{(-t,n)}{t^n}\right| \leq \frac{(r,n)}{r^n}, \qquad n\in\mathbb{N}. \tag{4}$$

Hence the series is majorized on A by

$$\sum_{n=0}^{\infty}\frac{(r,n)}{n!}\left(\frac{r/2}{r}\right)^n = \left(1-\frac{1}{2}\right)^{-r} = 2^r. \tag{5}$$

The series in (3) converges uniformly on A by the Weierstrass M test, and we can take the limit term by term as $t \to \infty$. For $n \geq 1$,

$$\frac{(-t, n)}{(-t)^n} = \left(1 - \frac{1}{t}\right)\left(1 - \frac{2}{t}\right) \cdots \left(1 - \frac{n-1}{t}\right) \to 1,$$

and thus

$$\lim_{t \to \infty} \left(1 + \frac{x}{t}\right)^t = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

Furthermore, (3) and (5) show that

$$\left|\left(1 + \frac{x}{t}\right)^t\right| \leq 2^r, \qquad 2|x| \leq r \leq |t|. \tag{6}$$

To prove (2), given any fixed $z \in \mathbb{C}^k$, we choose $r \geq 2 \max_{i=1}^k |z_i|$ and define A as we have done previously. The convex combination $u \cdot z$ satisfies $2|u \cdot z| \leq r$ for every $u' = (u_1, \ldots, u_{k-1})$ in the standard simplex $E \subset \mathbb{R}^{k-1}$. Hence (6) implies

$$\left|\left(1 + \frac{u \cdot z}{t}\right)^t\right| \leq 2^r, \qquad (u', t) \in E \times A. \tag{7}$$

Using this inequality and Lebesgue's theorem of dominated convergence (Appendix B.1 with Z, a, T chosen to be A, ∞, E), we evaluate

$$\lim_{t \to \infty} R_t\left(b, 1 + \frac{z}{t}\right) = \lim_{t \to \infty} \int_E \left(1 + \frac{u \cdot z}{t}\right)^t d\mu_b(u)$$

by taking the limit under the integral sign. By (1) the result is

$$\lim_{t \to \infty} R_t\left(b, 1 + \frac{z}{t}\right) = \int_E e^{u \cdot z} d\mu_b(u) = S(b, z). \qquad \square$$

A refinement of this theorem is useful. If we make slight changes in the preceding proof, it is easy to show (Exercise 5.10-2) that

$$e^x = \lim_{t \to \infty} \left[1 + \frac{\xi(t)}{t}\right]^t \qquad \text{if} \quad x = \lim_{t \to \infty} \xi(t), \tag{8}$$

$$S(b, z) = \lim_{t \to \infty} R_t\left[b, 1 + \frac{\zeta(t)}{t}\right] \qquad \text{if} \quad z = \lim_{t \to \infty} \zeta(t). \tag{9}$$

The k-tuple denoted by $1 + \zeta(t)/t$ has ith component $1 + \zeta_i(t)/t$.

Important examples of (9) are the expressions for the Bessel function J_0 as a limit of Legendre polynomials and J_ν as a limit of Gegenbauer polynomials (Exercise 6.7-13).

We can invert the confluent limiting process and return from the exponential function of x to the power function of x by integrating (see Exercise 3.2-3):

$$x^{-a} = \frac{1}{\Gamma(a)} \int_0^\infty y^{a-1} e^{-yx}\, dy, \qquad a, x \in \mathbb{C}_> . \tag{10}$$

The more general return from the S function to the R function is analogous, as shown in the following theorem.

THEOREM 5.10-2 Let $a \in \mathbb{C}_>$ and $b, z \in \mathbb{C}_>{}^k$. Then

$$R_{-a}(b, z) = \frac{1}{\Gamma(a)} \int_0^\infty y^{a-1} S(b, -yz)\, dy. \tag{11}$$

Proof Set $x = u \cdot z$ in (10) and integrate with respect to Dirichlet measure to obtain

$$\Gamma(a) R_{-a}(b, z) = \int_E \int_0^\infty y^{a-1} e^{-y(u \cdot z)}\, dy\, d\mu_b(u). \tag{12}$$

Define $r = \min\{\operatorname{Re} z_1, \ldots, \operatorname{Re} z_k\}$. Then $\operatorname{Re}(u \cdot z) \geq r > 0$ and

$$\begin{aligned} \int_E \int_0^\infty |y^{a-1} e^{-y(u \cdot z)}|\, dy\, d|\mu_b|(u) &\leq \int_E \int_0^\infty y^{\operatorname{Re} a - 1} e^{-yr}\, dy\, d|\mu_b|(u) \\ &\leq r^{-\operatorname{Re} a} \Gamma(\operatorname{Re} a) |\mu_b|(E) < \infty. \end{aligned}$$

In the last two steps we have used (10) and (4.4-4). It follows by Fubini's theorem (Appendix B.2) that the order of integration in (12) is immaterial. We integrate first over E, and obtain (11). □

A useful example of (11) is the Laplace transform of a Bessel function (Exercise 5.10-3). However, the most important aspect of (11) is the method of proof, which can be applied to various integral transforms of the type

$$g(x) = \int k(x, y) f(y)\, dy. \tag{13}$$

We set $x = u \cdot z$ and integrate with respect to Dirichlet measure. If a change in the order of integration can be justified [see Rognlie (1973)], the averaged transform is

$$G(b, z) = \int K(b, z; y) f(y)\, dy, \tag{14}$$

where the new kernel K is the Dirichlet average of the original kernel. For example, the S function in (11) is the Dirichlet average of the Laplace kernel e^{-yx} in (10). In the next section we shall consider the Dirichlet average of the Cauchy kernel.

5.11 Averages of Cauchy's integral formula

If f is an analytic function, we shall represent $F(b, z)$ by a contour integral which differs from Cauchy's integral formula only in that the Cauchy kernel is replaced by an R function. When we say that γ is a closed contour, we shall mean that it is a rectifiable Jordan curve. Its inner region will be denoted by $I(\gamma)$ with closure $\bar{I}(\gamma)$. The set of points lying on the curve (to be distinguished from the curve considered as a mapping of a real interval into the plane) will be denoted by $\gamma^* = \bar{I}(\gamma) - I(\gamma)$.

Let γ be a closed contour with positive orientation (e.g., described counterclockwise in the case of a circle), and let f be holomorphic on $I(\gamma)$ and continuous on $\bar{I}(\gamma)$. For every $\zeta \in I(\gamma)$ and every $n \in \mathbb{N}$, Cauchy's integral formula is

$$f^{(n)}(\zeta) = \frac{n!}{2\pi i} \int_\gamma f(s)(s - \zeta)^{-n-1}\, ds. \tag{1}$$

If $I(\gamma)$ contains the convex hull of $\{z_1, \ldots, z_k\}$, denoted by $\operatorname{con}(z)$, we may hope to obtain a simple generalization of (1) by taking Dirichlet averages. If we replace ζ by the convex combination $u \cdot z$ and integrate with respect to the Dirichlet measure μ_b, we obtain $F^{(n)}(b, z)$ on the left side. The kernel $(s - \zeta)^{-n-1}$ is replaced by $R_{-n-1}(b, s - z)$ on the right side, provided we can justify taking the average under the integral sign. This function is single-valued on γ, even though it may have branch points in $I(\gamma)$. For example, by (5.5-3),

$$R_{-1}(1, 1; s - x, s - y) = \frac{\log(s - x) - \log(s - y)}{y - x}, \qquad x \neq y.$$

This increases by $\pm 2\pi i/(y - x)$ if s encircles exactly one of the points x, y, but it returns to its initial value if s encircles the line segment joining x and y. The property of single-valuedness is asserted by the following theorem.

THEOREM 5.11-1 Let $n \in \mathbb{N}$, $b \in \mathbb{C}_>^k$, $z \in \mathbb{C}^k$, $s \in \mathbb{C}$, and $s - z = (s - z_1, \ldots, s - z_k)$. Let $\operatorname{con}(z)$ denote the convex hull of $\{z_1, \ldots, z_k\}$ and define $\mathbb{C}' = \mathbb{C} - \operatorname{con}(z)$. Then $R_{-n-1}(b, s - z)$ is holomorphic in (b, s) on $\mathbb{C}_>^k \times \mathbb{C}'$.

Proof Since it is holomorphic in b on $\mathbb{C}_>{}^k$ by Theorem 5.9-2, we need to show only that it is holomorphic in s on $\mathbb{C}'$. By (5.9-1),

$$R_{-n-1}(b, s - z) = \int_E (s - u \cdot z)^{-n-1}\, d\mu_b(u).$$

The integrand is holomorphic in s on $\mathbb{C}'$ (because n is an integer) and integrable for each $s \in \mathbb{C}'$. Let K be a compact set in $\mathbb{C}'$ and let δ be the distance between K and con(z). Then $\delta > 0$, and the integrand is bounded on K by δ^{-n-1}, which is independent of s and integrable with respect to $|\mu_b|$. It follows by Appendix B.4 that the integral is holomorphic in s on $\mathbb{C}'$. ▯

REPRESENTATION 5.11-2 Let $n \in \mathbb{N}$, $b \in \mathbb{C}_>{}^k$, $z \in \mathbb{C}^k$, and let γ be a rectifiable Jordan curve encircling the convex hull of $\{z_1, \ldots, z_k\}$ in the positive direction. Let f be holomorphic on $I(\gamma)$ and continuous on $\bar{I}(\gamma)$. Then

$$F^{(n)}(b, z) = \frac{n!}{2\pi i} \int_\gamma f(s) R_{-n-1}(b, s - z)\, ds, \tag{2}$$

where $s - z = (s - z_1, \ldots, s - z_k)$.

Remark If $z_1 = \cdots = z_k = \zeta$, (2) reduces to (1). The restriction on b will be removed in (6.3-4).

Proof We take the Dirichlet average of (1) and obtain

$$F^{(n)}(b, z) = \frac{n!}{2\pi i} \int_E \int_\gamma f(s)(s - u \cdot z)^{-n-1}\, ds\, d\mu_b(u). \tag{3}$$

Let L be the length of γ, let M be the maximum of $|f|$ on γ, and let δ be the distance between γ^* and con(z). Then L is finite because γ is rectifiable, M is finite because f is continuous on γ, and δ is strictly positive because γ^* and con(z) are compact. Moreover, by (4.4-4),

$$\int_E \int_\gamma |f(s)(s - u \cdot z)^{-n-1}|\,|ds|\, d|\mu_b|(u) \le M\delta^{-n-1} L |\mu_b|(E) < \infty.$$

It follows by Fubini's theorem (Appendix B.2) that the order of integration in (3) is immaterial. Carrying out the integration over E, we obtain (2). ▯

As an example of (2) take $f(s) = e^s$. If $b \in \mathbb{C}_>{}^k$, $z \in \mathbb{C}^k$, and con$(z) \subset I(\gamma)$, then

$$S(b, z) = \frac{1}{2\pi i} \int_\gamma e^s R_{-1}(b, s - z)\, ds. \tag{4}$$

More generally we find from (5.8-2) for every $n \in \mathbb{N}$,

$$S(b, z) = \frac{n!}{2\pi i} \int_\gamma e^s R_{-n-1}(b, s - z)\, ds. \tag{5}$$

5.12 Averages of $e^{1/x}$

The entire function e^x is said to have an essential singularity at ∞ because $e^{1/x}$ has an essential singularity at 0. Since $S(b, z)$ (Section 5.8) is the Dirichlet average of e^x, we expect its behavior at large $|z|$ to be related to the Dirichlet average of $e^{1/x}$, which we denote by T.

DEFINITION 5.12-1 Let $b \in \mathbb{C}_>^k$. Let H be a half-plane in $\mathbb{C} - \{0\}$ and let $z \in H^k$. Define

$$T(b, z) = \int_E e^{1/(u \cdot z)}\, d\mu_b(u). \tag{1}$$

The general case of this function does not seem to be of much practical importance, but throughout applied mathematics one encounters a limiting case with $k = 2$. We assume for the moment that x and y are real and negative, and we denote the limit of $T(\beta, \beta'; x, y)$ as $y \to 0$ by

$$T(\beta, \beta'; x, 0) = \frac{1}{\mathrm{B}(\beta, \beta')} \int_0^1 e^{1/ux} u^{\beta - 1}(1 - u)^{\beta' - 1}\, du. \tag{2}$$

Setting $1/u = 1 - tx$ and introducing the Euler measure (3.11-1),

$$d\lambda_\alpha(t) = \frac{1}{\Gamma(\alpha)} t^{\alpha - 1} e^{-t}\, dt,$$

we find

$$\frac{T(\beta, \beta'; x, 0)}{\Gamma(\beta + \beta')} = \frac{e^{1/x}(-x)^{\beta'}}{\Gamma(\beta)} \int_0^\infty (1 - tx)^{-\beta - \beta'}\, d\lambda_{\beta'}(t). \tag{3}$$

We shall see that this quantity is indeed related to the behavior of $S(\beta, \beta'; x, y)$ when $|x - y| \gg 1$. Its special cases include the error function, incomplete gamma function, exponential and logarithmic integrals, Hankel functions, Airy functions, parabolic cylinder functions, and irregular Coulomb wave functions. However, $T(\beta, \beta'; x, 0)$ is less convenient than the integral on the right side of (3) because of the following symmetry property.

THEOREM 5.12-2 Let $\alpha, \beta, -x \in \mathbb{C}_>$. Then

$$\int_0^\infty (1 - tx)^{-\alpha}\, d\lambda_\beta(t) = \int_0^\infty (1 - tx)^{-\beta}\, d\lambda_\alpha(t). \tag{4}$$

Proof By Exercise 3.2-3, for every $t \geq 0$,

$$(1 - tx)^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha - 1} e^{-(1 - tx)s}\, ds = \int_0^\infty e^{stx}\, d\lambda_\alpha(s),$$

and hence

$$\int_0^\infty (1 - tx)^{-\alpha} \, d\lambda_\beta(t) = \int_0^\infty \int_0^\infty e^{stx} \, d\lambda_\alpha(s) \, d\lambda_\beta(t). \tag{5}$$

The double integral converges absolutely because $|e^{stx}| \le 1$ and $|\lambda_\alpha|(\mathbb{R}_+) < \infty$ by (3.11-7). By Fubini's theorem (Appendix B.2) the order of integration is immaterial, and the integral is symmetric in α and β. $\square$

If we expand e^{stx} in Taylor series and formally integrate term by term without regard to convergence [cf. (3.11-5)], the double integral becomes the ${}_2F_0$ series $\sum (\alpha, n)(\beta, n)x^n/n!$. The Taylor series does not converge uniformly on the region of integration, the termwise integration is illegitimate, and the ${}_2F_0$ series diverges for every $x \neq 0$. Nevertheless the formal procedure explains why we shall use the symbol ${}_2F_0$ to designate not only the divergent series but also a well-defined function with integral representation (5). We shall see that the ${}_2F_0$ series is an asymptotic expansion of the ${}_2F_0$ function.

DEFINITION 5.12-3 Let α, β, $-x \in \mathbb{C}_>$, and let λ_α be an Euler measure (3.11-1). Define

$${}_2F_0(\alpha, \beta; x) = \int_0^\infty \int_0^\infty e^{stx} \, d\lambda_\alpha(s) \, d\lambda_\beta(t). \tag{6}$$

In the definition we have given preference to symmetry in α and β, even though the representation

$${}_2F_0(\alpha, \beta; x) = \int_0^\infty (1 - tx)^{-\alpha} \, d\lambda_\beta(t) \tag{7}$$

has a meaning for a larger domain in x and α. Analytic continuation in α, β, x will be effected by a different route which does not hide the symmetry. To allow values of α and β with negative real parts, we use the Taylor series with remainder (Exercise 5.8-3),

$$e^x = \sum_{m=0}^{p-1} \frac{x^m}{m!} + \frac{x^p}{p!} S(1, p; x, 0). \tag{8}$$

This holds even for $p = 0$ if we take the sum to be empty, since $S(1, 0; x, 0) = e^x$ by (5.8-7). If we substitute in (6) and use (3.11-3) and (3.11-4), we see that

$${}_2F_0(\alpha, \beta; x) = \sum_{m=0}^{p-1} \frac{(\alpha, m)(\beta, m)x^m}{m!} + \frac{(\alpha, p)(\beta, p)x^p}{p!} \int_0^\infty \int_0^\infty S(1, p; stx, 0) \, d\lambda_{\alpha+p}(s) \, d\lambda_{\beta+p}(t). \tag{9}$$

For any α, $\beta \in \mathbb{C}$ let p be the least nonnegative integer such that $\alpha + p$, $\beta + p \in \mathbb{C}_>$ and define ${}_2F_0$ by this equation. The definition coincides with (6) if α, $\beta \in \mathbb{C}_>$, which implies $p = 0$. We require p to be least only for the sake of definiteness. Since (8) holds for every $p \in \mathbb{N}$, (9) holds if p is replaced by any $n \in \mathbb{N}$ such that $\alpha + n$, $\beta + n \in \mathbb{C}_>$.

THEOREM 5.12-4 The function ${}_2F_0(\alpha, \beta; x)$ defined by (6) for $(\alpha, \beta, -x) \in \mathbb{C}_>{}^3$ has a holomorphic continuation in $(\alpha, \beta, -x)$ to $\mathbb{C}^2 \times \mathbb{C}_>$. If $n \in \mathbb{N}$ and $\alpha + n$, $\beta + n \in \mathbb{C}_>$, it is represented by

$$
\begin{aligned}
{}_2F_0(\alpha, \beta; x) &= \sum_{m=0}^{n-1} \frac{(\alpha, m)(\beta, m)x^m}{m!} \\
&\quad + \frac{(\alpha, n)(\beta, n)x^n}{n!} \int_0^\infty \int_0^\infty S(1, n; stx, 0)\, d\lambda_{\alpha+n}(s)\, d\lambda_{\beta+n}(t). \qquad (10)
\end{aligned}
$$

Proof Given $n \in \mathbb{N}$ we have already established (10) if $\alpha + n$, $\beta + n$, $-x \in \mathbb{C}_>$. Now we show that the integral

$$
\int_0^\infty \int_0^\infty S(1, n; stx, 0) s^{\alpha+n-1} t^{\beta+n-1} e^{-s-t}\, ds\, dt,
$$

is holomorphic. Since $\operatorname{Re}(stx) \le 0$, it is evident from (5.8-1) that $|S(1, n; stx, 0)| \le 1$. If K is a compact set in $\mathbb{C}_>$, we define $g_K(t)$ as in (3.3-1). If $\alpha + n$, $\beta + n \in K$ the integrand is majorized by $g_K(s)g_K(t)e^{-s-t}$, which is integrable. By Appendix B.4 it follows that the integral, and therefore ${}_2F_0$, is holomorphic in α, β, x on the domain defined by $\alpha + n$, $\beta + n$, $-x \in \mathbb{C}_>$. But n is arbitrarily large. □

If the parameters are real, we shall show that the error in approximating the ${}_2F_0$ function by a partial sum of the divergent ${}_2F_0$ series is less than the absolute value of the first omitted term.

THEOREM 5.12-5 Let $n \in \mathbb{N}$ and $\alpha + n$, $\beta + n \in \mathbb{R}_>$. Assume $|\operatorname{ph}(-x)| < \frac{1}{2}\pi$. Define u_m for every $m \in \mathbb{N}$ and r_n by

$$
u_m(x) = \frac{(\alpha, m)(\beta, m)x^m}{m!}, \qquad {}_2F_0(\alpha, \beta; x) = \sum_{m=0}^{n-1} u_m(x) + r_n(x). \qquad (11)
$$

Then $|r_n(x)| \le |u_n(x)|$. If it is further assumed that $x < 0$, then r_n and u_n are real and have the same sign.

Remark The next theorem shows that ${}_2F_0$ has an analytic continuation to the sector $|\operatorname{ph}(-x)| < 3\pi/2$. By continuity the present theorem still holds if $|\operatorname{ph}(-x)| \le \frac{1}{2}\pi$. It holds also if $\alpha + n$ or $\beta + n$ is 0, since the ${}_2F_0$ series terminates and $r_n = u_n$.

Proof By (5.8-1), $|S(1, n; stx, 0)| \le 1$ if $|\text{ph}(-x)| < \frac{1}{2}\pi$, and $0 < S(1, n; stx, 0) \le 1$ if $x < 0$. By (3.11-2) the integral in (10) is less than 1 in absolute value if $|\text{ph}(-x)| < \frac{1}{2}\pi$ and is positive if $x < 0$. ▯

It is necessary to require $-x \in \mathbb{C}_>$ in (10) so that $\text{Re}(-stx) \ge 0$. Even if $\frac{1}{2}\pi \le \text{ph}(-x) < \pi$, the same result can be achieved by rotating the path of integration in the s plane so that ph $s < 0$ and $\text{ph}(-sx) < \frac{1}{2}\pi$. The angle of rotation must not exceed $\frac{1}{2}\pi$ because of the factor e^{-s} in $d\lambda_{\alpha+n}(s)$. This allows $\text{ph}(-x)$ to approach π, and another rotation in the t plane allows $\text{ph}(-x)$ to approach $3\pi/2$. With the aid of rotations also in the opposite direction, we shall obtain an analytic continuation of ${}_2F_0(\alpha, \beta; x)$ to the sector $|\text{ph}(-x)| < 3\pi/2$. If $x \in \mathbb{C}_>$ this sector embraces two values of $\text{ph}(-x)$ which in general correspond to different values of ${}_2F_0$. A single-valued branch can be chosen by making a cut along the positive real axis of the x plane so that $|\text{ph}(-x)| < \pi$. Instead of doing so, we shall show that both branches have the same asymptotic expansion.

THEOREM 5.12-6 The function ${}_2F_0(\alpha, \beta; x)$ defined by (6) for $(\alpha, \beta, -x) \in \mathbb{C}_>{}^3$ has an analytic continuation to $\mathbb{C}^2 \times W$, where W is the sector $|\text{ph}(-x)| < 3\pi/2$. Define $\varphi = -(\frac{1}{3})\text{ph}(-x)$ and let $\gamma(\varphi)$ be a ray from 0 to ∞ at angle φ from the positive real axis. If $n \in \mathbb{N}$ and $\alpha + n$, $\beta + n \in \mathbb{C}_>$, then ${}_2F_0$ is represented by

$$ {}_2F_0(\alpha, \beta; x) = \sum_{m=0}^{n-1} \frac{(\alpha, m)(\beta, m)x^m}{m!} + \frac{(\alpha, n)(\beta, n)x^n}{n!} \int_{\gamma(\varphi)} \int_{\gamma(\varphi)} S(1, n; stx, 0)\, d\lambda_{\alpha+n}(s)\, d\lambda_{\beta+n}(t). \tag{12} $$

Remark The continuation is single-valued if $|\text{ph}(-x)| < \pi$.

Proof Consider

$$ \int_{\gamma(\theta)} \int_{\gamma(\theta)} S(1, n; stx, 0) s^{\alpha+n-1} t^{\beta+n-1} e^{-s-t}\, ds\, dt, \tag{13} $$

where θ is real and independent of φ. Because of the factor e^{-s-t} we require $|\theta| < \frac{1}{2}\pi$. Since $\text{ph}(-stx) = 2\theta - 3\varphi$, we require also $|2\theta - 3\varphi| \le \frac{1}{2}\pi$ so that $\text{Re}(stx) \le 0$ and $|S(1, n; stx, 0)| \le 1$. Thus the integral converges absolutely on the set (see Fig. 5.12-1)

$$ D = \{(\theta, \varphi) \in \mathbb{R}^2 : |\theta| < \tfrac{1}{2}\pi,\ |2\theta - 3\varphi| \le \tfrac{1}{2}\pi\}. $$

To show analyticity in α and β, we set $s = \sigma e^{i\theta}$ and $t = \tau e^{i\theta}$, whence

$$ |s^{\alpha+n-1} e^{-s}\, ds| = e^{-\theta\, \text{Im}\, \alpha} \sigma^{\text{Re}\, \alpha + n - 1} e^{-\sigma \cos\theta}\, d\sigma. $$

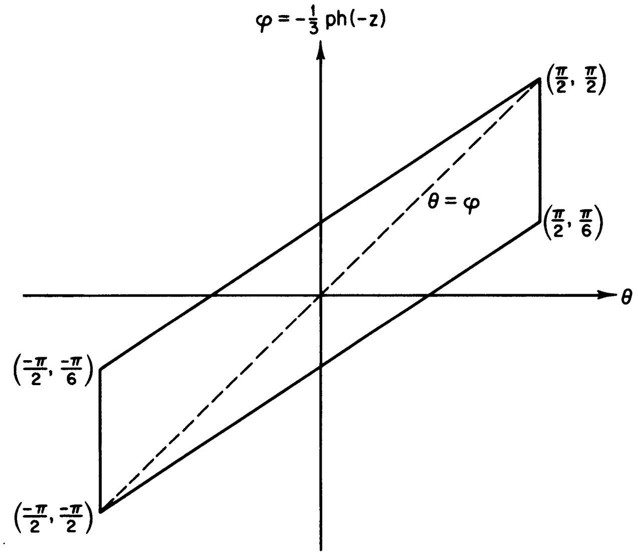

Figure 5.12-1 The parallelogram is the boundary of the set D, and the dashed line is the diagonal $\theta = \varphi$.

If K is a compact set in $\mathbb{C}_>$, we define the function g_K as in (3.3-1). If $\alpha + n$, $\beta + n \in K$ the integrand is majorized by a constant multiple of

$$g_K(\sigma)g_K(\tau)\exp[-(\sigma + \tau)\cos\theta]\, d\sigma\, d\tau,$$

which is integrable on $\mathbb{R}_+{}^2$. By Appendix B.4 it follows that the integral (13) is holomorphic in α and β if $\alpha + n$, $\beta + n \in \mathbb{C}_>$.

We turn now to analyticity in x. On a section of D at fixed φ the integral is independent of θ by Cauchy's integral theorem. A section of D at fixed θ determines a half-plane in the x plane on which the integral is holomorphic by Appendix B.4 since $|S(1,\, n;\, stx,\, 0)| \leq 1$. Two values of θ differing by less than $\frac{1}{2}\pi$ correspond to overlapping half-planes and therefore to holomorphic functions which are analytic continuations of one another. In (12) the choice $\theta = \varphi$ selects the diagonal of D and is merely a convenient way of admitting all φ with $|\varphi| < \frac{1}{2}\pi$.

Thus ${}_2F_0(\alpha, \beta; x)$ is continued analytically to the sector $|\mathrm{ph}(-x)| < 3\pi/2$ For every x in this sector it is entire in α and β because it is holomorphic if $\alpha + n$, $\beta + n \in \mathbb{C}_>$ where n is arbitrarily large. □

We shall prove next that the ${}_2F_0$ series is an asymptotic expansion of the ${}_2F_0$ function as $x \to 0$ in the sector $|\mathrm{ph}(-x)| < 3\pi/2$. This sector includes real values of x of both signs. We shall obtain a bound for the error made in approximating the ${}_2F_0$ function by a partial sum of the ${}_2F_0$ series.

THEOREM 5.12-7 Let $n \in \mathbb{N}$, $x \in \mathbb{C}$, and $\alpha + n$, $\beta + n \in \mathbb{C}_>$. Assume $|\text{ph}(-x)| < 3\pi/2$ and define $\varphi = -\frac{1}{3}\,\text{ph}(-x)$. Then

$$ {}_2F_0(\alpha, \beta; x) = \sum_{m=0}^{n-1} \frac{(\alpha, m)(\beta, m)x^m}{m!} + r_n, \tag{14}$$

$$ |r_n| \le \frac{|x|^n}{n!} \frac{\Gamma(\text{Re}\,\alpha + n)\Gamma(\text{Re}\,\beta + n)}{|\Gamma(\alpha)\Gamma(\beta)|} \frac{e^{-\varphi\,\text{Im}(\alpha+\beta)}}{(\cos\varphi)^{\text{Re}(\alpha+\beta)+2n}}. \tag{15}$$

Assume further that $|\text{ph}(-x)| \le \pi$ and $\alpha + n - \frac{1}{2}$, $\beta + n - \frac{1}{2} \in \mathbb{C}_>$. Then

$$ |r_n| \le \left| \frac{(\alpha, n)(\beta, n)}{n!} x^n \right| 2^{\text{Re}(\alpha+\beta)+2n} e^{\pi|\text{Im}\,\alpha| + \pi|\text{Im}\,\beta|}. \tag{16}$$

Remark The leading factor on the right side of (16) is the first neglected term in the ${}_2F_0$ series. An inequality which is sharper but less simple than (16) can be obtained by using (15) and (3.10-7).

Proof In (12) the remainder is majorized by

$$ \left| \frac{(\alpha, n)(\beta, n)}{n!} x^n \right| \int_{\gamma(\varphi)} d|\lambda_{\alpha+n}|(s) \int_{\gamma(\varphi)} d|\lambda_{\beta+n}|(t). $$

Use (3.11-8) to prove (15); use (3.11-9) to prove (16) and note that $|\varphi| \le \pi/3$. ▯

Since (15) shows that $r_n \to 0$ as $x \to 0$ in the sector $|\text{ph}(-x)| \le 3\pi/2 - \delta$, where δ is any positive number, the ${}_2F_0$ series is an asymptotic expansion of the ${}_2F_0$ function in this sector:

$$ {}_2F_0(\alpha, \beta; x) \sim \sum_{m=0}^{\infty} \frac{(\alpha, m)(\beta, m)}{m!} x^m. \tag{17}$$

Even though the series diverges for every $x \neq 0$, its partial sums are useful approximations to the function for sufficiently small $|x|$. The error bound (15) becomes large when $\text{ph}(-x)$ is close to $\pm 3\pi/2$ because of the positive power of $\cos\varphi$ in the denominator. Unless an error bound is calculated, the partial sums of an asymptotic series are riskier approximations than those of a convergent series because the size of successive terms is often no clue to the size of the error.

It may seem paradoxical that the partial sums are always single-valued while the ${}_2F_0$ function is multiple-valued in the sector $|\text{ph}(-x)| < 3\pi/2$.

As $x \to 0$, however, the difference between the principal branch with $|\text{ph}(-x)| < \pi$ and a branch with $\pi < |\text{ph}(-x)| < 3\pi/2$ tends to 0 exponentially and therefore faster than any power of x. The error bound (15) holds for both branches, even though it is proportional to an arbitrarily large power of $|x|$.

We shall now connect ${}_2F_0$ with the S function in two variables. The asymptotic properties of $S(\beta, \beta'; x, y)$ as $|x - y| \to \infty$ will then be deducible from Theorem 5.12-7.

THEOREM 5.12-8 Let $x, y \in \mathbb{C}$ and $\beta, \beta' \in \mathbb{C}_>$. Assume $x \neq y$, $|\text{ph}(x - y)| < 3\pi/2$, and $|\text{ph}(y - x)| < 3\pi/2$. Then

$$\frac{S(\beta, \beta'; x, y)}{\Gamma(\beta + \beta')} = \frac{e^x(x - y)^{-\beta'}}{\Gamma(\beta)}\, {}_2F_0\left(1 - \beta, \beta'; \frac{1}{x - y}\right) + \frac{e^y(y - x)^{-\beta}}{\Gamma(\beta')}\, {}_2F_0\left(1 - \beta', \beta; \frac{1}{y - x}\right). \tag{18}$$

Remark The right side is entire in β and β', and we shall find in Corollary 6.3-3 that the left side is too. By the permanence of functional relations the restriction on β and β' can then be dropped. Provided that the stated conditions on the phases are satisfied, the right side of (18) is independent of whether $\text{ph}(x - y) - \text{ph}(y - x)$ is π or $-\pi$. See Section 7.9 for a corollary of this theorem and two applications to wave mechanics.

Proof By (5.8-1),

$$\frac{S(\beta, \beta'; x, y)}{\Gamma(\beta + \beta')} = \frac{1}{\Gamma(\beta)\Gamma(\beta')} \int_0^1 e^{ux + (1-u)y} u^{\beta - 1}(1 - u)^{\beta' - 1}\, du. \tag{19}$$

Since S is entire in x and y, Theorem 5.12-6 and the permanence of functional relations make it sufficient to prove (18) when $0 < |\text{ph}(y - x)| < \pi$. Let γ be a ray in the u plane from 0 to ∞ at angle $-\text{ph}(y - x)$ from the positive real axis. If u is on γ and $u \neq 0$, then $u(x - y)$ is real and negative because

$$\text{ph}[u(x - y)] = \text{ph}\, u + \text{ph}(x - y) = \text{ph}(x - y) - \text{ph}(y - x) = \pm\pi.$$

Hence $\exp[ux + (1 - u)y] \to 0$ as $u \to \infty$ on γ. Let γ' be the ray from 1 to ∞ parallel to γ. By Cauchy's integral theorem we can set $\int_0^1 = \int_\gamma - \int_{\gamma'}$. In the last integral replacement of u by $1 + u$ changes the path of integration from γ' to γ. Thus the integral in (19) equals

$$\int_\gamma e^{ux + (1-u)y} u^{\beta - 1}(1 - u)^{\beta' - 1}\, du - \int_\gamma e^{(1+u)x - uy}(1 + u)^{\beta - 1}(-u)^{\beta' - 1}\, du.$$

Setting $u(y-'x)=t$ and using (7) changes this difference into

$$e^y(y-x)^{-\beta}\int_0^\infty e^{-t}t^{\beta-1}\left(1-\frac{t}{y-x}\right)^{\beta'-1}dt$$

$$+\,e^x(x-y)^{-\beta'}\int_0^\infty e^{-t}\left(1-\frac{t}{x-y}\right)^{\beta-1}t^{\beta'-1}\,dt$$

$$=e^y(y-x)^{-\beta}\Gamma(\beta)\,{}_2F_0\left(1-\beta',\beta;\frac{1}{y-x}\right)$$

$$+\,e^x(x-y)^{-\beta'}\Gamma(\beta')\,{}_2F_0\left(1-\beta,\beta';\frac{1}{x-y}\right).$$

Substitution in (19) completes the proof of (18). ▯

THEOREM 5.12-9 Let $\alpha,\beta\in\mathbb{C}$ and $x\in\mathbb{C}-\{0\}$. Assume $|\mathrm{ph}\,x|<3\pi/2$ and $\alpha-\beta\notin\mathbb{Z}$. Assume further that $0<\operatorname{Re}\alpha<1$ and $0<\operatorname{Re}\beta<1$ (see the Remark below). Then

$$\begin{aligned}{}_2F_0(\alpha,\beta;-1/x)=\frac{\Gamma(\beta-\alpha)}{\Gamma(\beta)}x^\alpha S(\alpha,1-\beta;x,0)\\+\frac{\Gamma(\alpha-\beta)}{\Gamma(\alpha)}x^\beta S(\beta,1-\alpha;x,0).\end{aligned}\tag{20}$$

Remark The restriction on $\operatorname{Re}\alpha$ and $\operatorname{Re}\beta$ can be dropped after the S function has been continued analytically in the parameters in Section 6.3.

Proof Substitute (18) for each S function on the right side of (20). The resulting expression has two terms containing ${}_2F_0(1-\alpha,1-\beta;1/x)$, and the sum of the two is proportional to

$$\Gamma(\beta-\alpha)\Gamma(1+\alpha-\beta)+\Gamma(\alpha-\beta)\Gamma(1+\beta-\alpha)=0.$$

There are also two terms containing ${}_2F_0(\alpha,\beta;-1/x)$, and the sum of their coefficients is

$$\frac{\Gamma(\beta-\alpha)\Gamma(1+\alpha-\beta)}{\Gamma(\beta)\Gamma(1-\beta)}x^\alpha(-x)^{-\alpha}+\frac{\Gamma(\alpha-\beta)\Gamma(1+\beta-\alpha)}{\Gamma(\alpha)\Gamma(1-\alpha)}x^\beta(-x)^{-\beta}$$

$$=\frac{1}{\sin\pi(\alpha-\beta)}(-e^{\pm i\pi\alpha}\sin\pi\beta+e^{\pm i\pi\beta}\sin\pi\alpha)$$

$$=\frac{1}{\sin\pi(\alpha-\beta)}(-\cos\pi\alpha\sin\pi\beta+\cos\pi\beta\sin\pi\alpha)$$

$$=1.$$

We have used (3.9-3) and the relation $\mathrm{ph}\,x-\mathrm{ph}(-x)=\pm\pi$. ▯

Since the S function is entire, (20) exhibits clearly the many-valued character of ${}_2F_0$ and can be used to continue it analytically to sectors with $|\text{ph}\, x| \geq 3\pi/2$. For this reason (20) is sometimes used, originally by Erdélyi (1939), to define ${}_2F_0$. Cases with $\alpha - \beta \in \mathbb{Z}$ require taking a limit of the right side.

Since the S function satisfies the confluent hypergeometric equation (5.8-12), it follows from (20) that ${}_2F_0$ will be met in solving differential equations of the type (5.8-14). Solutions known as Whittaker functions are related to S and ${}_2F_0$ by

$$\begin{aligned} M_{k,m}(x) &= x^{m+1/2} S(\tfrac{1}{2} - k + m, \tfrac{1}{2} + k + m; \tfrac{1}{2}x, -\tfrac{1}{2}x), \\ W_{k,m}(x) &= e^{-x/2} x^k \,{}_2F_0(\tfrac{1}{2} - k + m, \tfrac{1}{2} - k - m; -1/x). \end{aligned} \tag{21}$$

Alternative choices made by Tricomi are

$$\begin{aligned} \Phi(a, c; x) &= S(a, c - a; x, 0),. \\ \Psi(a, c; x) &= x^{-a}\,{}_2F_0(a, 1 + a - c; -1/x). \end{aligned} \tag{22}$$

The letters M and U are often used in place of Φ and Ψ, respectively. Whittaker functions and Tricomi functions were chosen to be solutions of standard differential equations, whereas S and ${}_2F_0$ were chosen for permutation symmetry. Whittaker's $M_{k,m}$ has in general a branch point at $x = 0$ while Φ and S are entire.

In Exercises 5.12-1 to 5.12-4 the error functions, incomplete gamma functions, exponential integral, and logarithmic integral are expressed in terms of S and ${}_2F_0$. The parabolic cylinder functions are

$$D_\nu(x) = e^{-(x^2)/4} x^\nu \,{}_2F_0\left(\frac{-\nu}{2}, \frac{1-\nu}{2}; \frac{-2}{x^2}\right). \tag{23}$$

Macdonald's function (the second kind of modified Bessel function) is

$$K_\nu(x) = e^{-x}(\pi/2x)^{1/2}\,{}_2F_0(\tfrac{1}{2} + \nu, \tfrac{1}{2} - \nu; -1/2x), \tag{24}$$

and the Hankel functions or Bessel functions of the third kind are

$$\begin{aligned} H_\nu^{(1)}(x) &= i^{-\nu-1/2} e^{ix}(2/\pi x)^{1/2}\,{}_2F_0(\tfrac{1}{2} + \nu, \tfrac{1}{2} - \nu; 1/2ix), \\ H_\nu^{(2)}(x) &= i^{\nu+1/2} e^{-ix}(2/\pi x)^{1/2}\,{}_2F_0(\tfrac{1}{2} + \nu, \tfrac{1}{2} - \nu; -1/2ix). \end{aligned} \tag{25}$$

Weber's function, sometimes called Neumann's function or the Bessel function of the second kind, is

$$Y_\nu(x) = N_\nu(x) = \frac{1}{2i}[H_\nu^{(1)}(x) - H_\nu^{(2)}(x)], \tag{26}$$

and it follows from (5.8-23), (18), and (25) that

$$J_\nu(x) = \tfrac{1}{2}[H_\nu^{(1)}(x) + H_\nu^{(2)}(x)]. \tag{27}$$

Asymptotic expansions of all these functions can be obtained from (14). In particular, that for $H_\nu^{(1)}(x)$ holds in the sector $|\mathrm{ph}(ix)| < 3\pi/2$ or $-\pi < \mathrm{ph}\, x < 2\pi$, while that for $H_\nu^{(2)}(x)$ holds if $-2\pi < \mathrm{ph}\, x < \pi$. As $x \to \infty$ in the sector $|\mathrm{ph}\, x| < \pi$, where both are valid, we find

$$\begin{aligned} Y_\nu(x) &\sim (2/\pi x)^{1/2} \sin(x - \tfrac{1}{2}\nu\pi - \pi/4), \\ J_\nu(x) &\sim (2/\pi x)^{1/2} \cos(x - \tfrac{1}{2}\nu\pi - \pi/4). \end{aligned} \tag{28}$$

These asymptotic approximations preserve the evident analogy between (26) and (27) and the formulas for $\sin x$ and $\cos x$ in terms of $e^{\pm ix}$.

NOTES

Dirichlet averages are discussed by Carlson (1969, 1970c), the latter reference emphasizing effects of permutation symmetry. For further properties of the R function see later chapters and Carlson (1963). Fractional integration and its applications to axially symmetric potential theory are treated by Erdélyi (1965), Sneddon (1966), Oldham (1974), and Lavoie (1976). Information about the Euler–Poisson–Darboux equation can be found in Darboux (1894, Vol. 2, pp. 54–70) and Courant (1953, Vol. 2, pp. 646–647, 699–703); the latter reference discusses a generalized form of the equation and its relation to wave propagation. Sharper error bounds than those in (5.12-15) are given by Olver (1965) and Carlson (1978).

FORMULARY

$$F(b_1, \ldots, b_k; x, \ldots, x) = f(x), \tag{5.2-2}$$

$$F^{(n)}(b, z) = \int_E f^{(n)}(u \cdot z)\, d\mu_b(u), \tag{5.3-1}$$

$$\left(\sum_{i=1}^{k} D_i\right)^n F = F^{(n)}, \tag{5.3-3}$$

$$v = F(b, z), \qquad (z_i - z_j)D_i D_j v + b_i D_j v - b_j D_i v = 0, \qquad i, j = 1, \ldots, k, \tag{5.4-2}$$

$$F(1, 1; x, y) = \frac{1}{x - y}\int_y^x f(t)\, dt, \tag{5.5-2}$$

$$f[x, y] = \frac{f(x) - f(y)}{x - y} = F'(1, 1; x, y), \tag{5.5-3}$$

$$f(x) = \sum_{n=0}^{p-1} \frac{1}{n!} f^{(n)}(a)(x - a)^n + \frac{1}{p!} F^{(p)}(1, p; x, a)(x - a)^p, \tag{5.5-8}$$

$$F(b, z) = \sum_{i=1}^{k} w_i F(b + e_i, z), \qquad w_i = b_i/c, \tag{5.6-4}$$

$$D_i F(b, z) = w_i F'(b + e_i, z), \tag{5.6-5}$$

$$(z_i - z_j)F'(b, z) + (c - 1)[F(b - e_i, z) - F(b - e_j, z)] = 0, \tag{5.6-10}$$

$$F(b, z) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(\lambda) R_n(b, z - \lambda), \tag{5.7-4}$$

$$S(b, z) = \int_E e^{u \cdot z}\, d\mu_b(u), \tag{5.8-1}$$

$$S^{(n)} = S, \tag{5.8-2}$$

$$S(b, z + \lambda) = e^{\lambda} S(b, z), \tag{5.8-3}$$

$$S(b, z) = \sum_{n=0}^{\infty} \frac{1}{n!} R_n(b, z), \tag{5.8-4}$$

$$S(\beta, \gamma - \beta; x, y) = e^{y}\, {}_1F_1(\beta; \gamma; x - y), \qquad {}_1F_1(\beta; \gamma; x) = S(\beta, \gamma - \beta; x, 0), \tag{5.8-6}$$

$$J_\mu(x) = \frac{(x/2)^\mu}{\Gamma(1 + \mu)} S\left(\frac{1}{2} + \mu, \frac{1}{2} + \mu; ix, -ix\right), \qquad j_L(x) = (\pi/2x)^{1/2} J_{L+1/2}(x). \tag{5.8-23}$$

$$R_t(b, z) = \int_E (u \cdot z)^t\, d\mu_b(u), \tag{5.9-1}$$

$$\sum_{i=1}^{k} D_i R_t = t R_{t-1}, \qquad \sum_{i=1}^{k} z_i D_i R_t = t R_t, \tag{5.9-2}$$

$$R_t(b, \lambda z) = \lambda^t R_t(b, z), \tag{5.9-3}$$

$$R_{-a}(b, 1 - z) = \sum_{n=0}^{\infty} \frac{(a, n)}{n!} R_n(b, z), \qquad |z_i| < 1, \quad i = 1, \ldots, k, \tag{5.9-4}$$

$$R_t(b, z) = \sum_{i=1}^{k} w_i R_t(b + e_i, z), \tag{5.9-5}$$

$$R_{t+1}(b, z) = \sum_{i=1}^{k} w_i z_i R_t(b + e_i, z). \tag{5.9-6}$$

$$c R_t(b, z) = (c + t) R_t(b + e_i, z) - t z_i R_{t-1}(b + e_i, z), \qquad i = 1, \ldots, k, \tag{5.9-7}$$

$$D_i R_t = w_i t R_{t-1}(b + e_i, z), \tag{5.9-9}$$

$$(z_i D_i + b_i) R_t(b, z) = w_i (c + t) R_t(b + e_i, z), \tag{5.9-10}$$

$$R_{-\alpha}(\beta, \gamma - \beta; x, y) = y^{-\alpha}\, {}_2F_1(\alpha, \beta; \gamma; 1 - x/y), \tag{5.9-11}$$

$${}_2F_1(\alpha, \beta; \gamma; x) = R_{-\alpha}(\beta, \gamma - \beta; 1 - x, 1), \tag{5.9-12}$$

$$R_{-\alpha}(\beta, \beta'; x, y) = y^{\beta - \alpha} R_{-\beta}(\alpha, \alpha'; x, y), \tag{5.9-19}$$

$$R_{-\alpha}(\beta, \beta'; x, y) = x^{\beta' - \alpha} R_{-\beta'}(\alpha', \alpha; x, y), \tag{5.9-20}$$

$$R_{-\alpha}(\beta, \beta'; x, y) = x^{\beta' - \alpha} y^{\beta - \alpha} R_{-\alpha'}(\beta', \beta; x, y), \tag{5.9-21}$$

$$R_{-\beta-\beta'}(\beta, \beta'; x, y) = x^{-\beta} y^{-\beta'}. \tag{5.9-22}$$

The next two equations hold only for the case of two variables.

$$w_i(z_j - z_i) R_t(b + e_i, z) = z_j R_t(b, z) - R_{t+1}(b, z) \qquad (i = 1, 2, \quad j = 3 - i), \tag{5.9-24}$$

$$(c+t)R_{t+1}(b,z) - \sum_{i=1}^{2}(b_i+t)z_i R_t(b,z) + tz_1z_2R_{t-1}(b,z) = 0, \tag{5.9-25}$$

$$S(b,z) = \lim_{t\to\infty} R_t(b, 1+z/t), \tag{5.10-2}$$

$$F^{(n)}(b,z) = \frac{n!}{2\pi i}\int_\gamma f(s)R_{-n-1}(b, s-z)\,ds, \tag{5.11-2}$$

$$\frac{S(\beta,\beta';x,y)}{\Gamma(\beta+\beta')} = \frac{e^x(x-y)^{-\beta'}}{\Gamma(\beta)}\,{}_2F_0\left(1-\beta,\beta';\frac{1}{x-y}\right) + (x \rightleftarrows y, \beta \rightleftarrows \beta'), \tag{5.12-18}$$

$${}_2F_0(\alpha,\beta;-1/x) = \frac{\Gamma(\beta-\alpha)}{\Gamma(\beta)}x^\alpha S(\alpha, 1-\beta; x, 0) + (a \rightleftarrows \beta). \tag{5.12-20}$$

EXERCISES

5.1-1 Let $\beta, \beta' \in \mathbb{C}_>$ and let f be piecewise continuous on the line segment in $\mathbb{C}$ with endpoints x, y. Show that

$$B(\beta,\beta')(x-y)^{\beta+\beta'-1}F(\beta,\beta';x,y) = \int_y^x f(t)(t-y)^{\beta-1}(x-t)^{\beta'-1}\,dt,$$

where the path of integration is the line segment joining x and y.

5.1-2 Let $\beta, \beta' \in \mathbb{C}_>$ and let f be piecewise continuous on the line segment in $\mathbb{C}$ with endpoints $x+y$ and $x-y$. Show that

$$\begin{aligned} B(\beta,\beta')F(\beta,\beta';x+y,x-y) &= 2\int_0^{\pi/2} f(x+y\cos 2\theta)(\cos\theta)^{2\beta-1}(\sin\theta)^{2\beta'-1}\,d\theta \\ &= 2^{1-\beta-\beta'}\int_{-1}^{1} f(x+yt)(1+t)^{\beta-1}(1-t)^{\beta'-1}\,dt. \end{aligned}$$

5.1-3 Let $\beta \in \mathbb{C}_>$, and let f be piecewise continuous on the line segment in $\mathbb{C}$ with endpoints $x+y$ and $x-y$. Show that

$$B(\beta,\tfrac{1}{2})F(\beta,\beta;x+y,x-y) = \int_0^\pi f(x+y\cos\varphi)(\sin\varphi)^{2\beta-1}\,d\varphi = \int_{-1}^{1} f(x+yt)(1-t^2)^{\beta-1}\,dt.$$

5.1-4 Let $x, y \in \mathbb{C}$ and $\beta \in \mathbb{C}_>$. Let f be holomorphic on an open disk with centre x and radius $R > |y|$. If $0 < \theta < \pi$, show that

$$B(\beta,\tfrac{1}{2})F(\beta,\beta;x+ye^{i\theta},x+ye^{-i\theta}) = 2^{\beta-1}(\sin\theta)^{1-2\beta}\int_{-\theta}^{\theta} f(x+ye^{i\varphi})e^{i\beta\varphi}(\cos\varphi-\cos\theta)^{\beta-1}\,d\varphi.$$

5.2-1 Let the real-valued function f be convex on an interval I in $\mathbb{R}$. Assume $b \in \mathbb{R}_>^{\,k}$ and define the weights w by (5.6-2). Prove that

$$f\left(\sum_{i=1}^{k} w_i x_i\right) \le F(b,x) \le \sum_{i=1}^{k} w_i f(x_i), \qquad x \in I^k,$$

with reversed inequalities if f is concave instead of convex.

5.2-2 If $b \in \mathbb{C}_>{}^k$ and $z \in \mathbb{C}^k$, show that

$$|F(b, z)| \le \frac{\mathrm{B}(\operatorname{Re} b)}{|\mathrm{B}(b)|} \sup |f(\zeta)|,$$

where the supremum is taken over all $\zeta \in \operatorname{con}(z)$.

5.3-1 Use the assumptions of Definition 5.3-1 and let $x + z = (x + z_1, \ldots, x + z_k) \in \Omega^k$. Show that

$$\frac{d^n}{dx^n} F(b, x + z) = F^{(n)}(b, x + z).$$

5.4-1 Let f be holomorphic on an open convex set Ω in $\mathbb{C}$. If $x, y \in \Omega$ and $\beta \in \mathbb{C}_>$, prove that

$$\left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right) F(\beta, \beta; x, y) = \frac{x - y}{\beta} \frac{\partial^2}{\partial x\, \partial y} F(\beta, \beta; x, y) = \frac{x - y}{2(2\beta + 1)} F''(\beta + 1, \beta + 1; x, y).$$

5.4-2 Let f be holomorphic on an open convex set Ω in $\mathbb{C}$. If $x, y \in \Omega$ and $\beta \in \mathbb{C}_>$, prove that

$$(x - y)^2 F''(\beta + 2, \beta + 2; x, y) = 4(2\beta + 1)(2\beta + 3)[F(\beta, \beta; x, y) - F(\beta + 1, \beta + 1; x, y)].$$

For an important special case see Exercise 5.8-2. See also Exercise 5.4-8.

5.4-3 Let f be holomorphic on an open convex set Ω in $\mathbb{C}$. If $x, y \in \Omega$, $\alpha \in \mathbb{C}$, and $\beta \in \mathbb{C}_>$, prove that

$$\begin{aligned}(x - y)^{1-\alpha}&\left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right)[(x - y)^\alpha F(\beta + 1, \beta + 1; x, y)] \\ &= 2(2\beta + 1)F(\beta, \beta; x, y) + 2(\alpha - 2\beta - 1)F(\beta + 1, \beta + 1; x, y).\end{aligned}$$

For an important special case see Exercise 5.8-8.

5.4-4 If f is twice continuously differentiable, show that $\varphi = (x - y)^{-\alpha} F(\beta, \beta'; x, y)$ satisfies

$$(x - y)\varphi_{xy} + (\alpha + \beta)\varphi_y - (\alpha + \beta')\varphi_x - \alpha(\alpha - 1 + \beta + \beta')(x - y)^{-1}\varphi = 0,$$

where $\varphi_x = \partial\varphi/\partial x$ and $\varphi_{xy} = \partial^2\varphi/\partial x\, \partial y$.

5.4-5 If f is twice continuously differentiable, show that $v = (x - y)^{1-\beta-\beta'} \cdot F(1 - \beta', 1 - \beta; x, y)$ is a second solution of (5.4-1). It is well defined if β and β' have real parts less than unity (see Section 6.3 for removal of this restriction), but it coincides with $F(\beta, \beta'; x, y)$ if $\beta + \beta' = 1$.

5.4-6 Let σ be the interior of a square in the x, t plane in which one diagonal is an open interval I of the x axis. Let $f \in C_2(I)$ and $\beta, \beta' \in \mathbb{C}_>$. For all $(x, t) \in \sigma$ define $v(x, t) = F(\beta, \beta'; x + t, x - t)$. Show that

$$t(v_{xx} - v_{tt}) + (\beta - \beta')v_x - (\beta + \beta')v_t = 0,$$

where $v_x = \partial v/\partial x$ and $v_{xx} = \partial^2 v/\partial x^2$. If $\beta = \beta'$, show further that v satisfies the Euler–Poisson–Darboux equation,

$$v_{xx} = v_{tt} + (2\beta/t)v_t,$$

with initial conditions $v(x, 0) = f(x)$ and $v_t(x, 0) = 0$, $x \in I$.

5.4-7 If $t \in \mathbb{C}$, show that $v_0 = (t + z_1)^{-b_1} \cdots (t + z_k)^{-b_k}$ is a solution of (5.4-2). Show that another solution is

$$v = \int_\gamma f(t) \prod_{i=1}^{k} (t + z_i)^{-b_i} dt,$$

where the function f and the b_i are assumed to be such that the integral exists and can be differentiated twice under the integral sign. The contour γ may be closed (on the Riemann surface of the integrand), or γ may be an open path with endpoints which are either independent of z or chosen from among the points $-z_1, \ldots, -z_k$.

5.4-8 Let f be holomorphic on an open convex set Ω in $\mathbb{C}$. Let $x, y \in \Omega$ and $\beta - 1, \beta' - 1 \in \mathbb{C}_>$, and define $c = \beta + \beta'$. Prove that

$$\frac{\beta\beta'(x-y)^2}{c^2(c^2-1)} F''(\beta+1, \beta'+1; x, y) + \frac{(\beta'-\beta)(x-y)}{c(c-2)} F'(\beta, \beta'; x, y)$$
$$= F(\beta-1, \beta'-1; x, y) - F(\beta, \beta'; x, y).$$

5.5-1 Let Ω be either an open interval in $\mathbb{R}$ or a convex open set in $\mathbb{C}$, and let f be twice continuously differentiable on Ω. If x, y, z are distinct points in Ω, show that

$$\tfrac{1}{2}F''(1, 1, 1; x, y, z) = \frac{f(x)}{(x-y)(x-z)} + \frac{f(y)}{(y-z)(y-x)} + \frac{f(z)}{(z-x)(z-y)}.$$

The right side is expressed as a ratio of determinants in Exercise 3.5-6. More generally let $n \in \mathbb{N}$ and let f be n times continuously differentiable on Ω. Let $z_1, \ldots, z_{n+1}$ be distinct points in Ω and define $p_{n+1}(t) = (t - z_1)\cdots(t - z_{n+1})$. Show that the nth divided difference of f is

$$\frac{1}{n!} F^{(n)}(1, \ldots, 1; z_1, \ldots, z_{n+1}) = \sum_{i=1}^{n+1} \frac{f(z_i)}{p'_{n+1}(z_i)}.$$

5.5-2 In the calculus of finite differences the forward difference operator Δ is defined by $\Delta f(x) = f(x+h) - f(x)$. If $n = 2, 3, \ldots,$ define $\Delta^n f = \Delta(\Delta^{n-1} f)$. For example the second forward difference is $\Delta^2 f(x) = f(x+2h) - 2f(x+h) + f(x)$. Let Ω and f satisfy the assumptions of Lemma 5.5-1, and suppose that $x + mh \in \Omega$, $m = 0, 1, \ldots, n$. Show that

$$\Delta^n f(x) = h^n F^{(n)}(1, \ldots, 1; x, x+h, \ldots, x+nh).$$

For an interesting special case see Exercise 5.8-10.

5.5-3 The central difference operator δ is defined by $\delta f(x) = f(x + \frac{1}{2}h) - f(x - \frac{1}{2}h)$. If $n = 2, 3, \ldots,$ define $\delta^n f = \delta(\delta^{n-1} f)$. For example the second central difference is $\delta^2 f(x) = f(x+h) - 2f(x) + f(x-h)$. Let Ω and f satisfy the assumptions of Lemma 5.5-1, and suppose that $x + mh/2 \in \Omega$ for $m = n, n-2, \ldots, -n$. Show that

$$\delta^n f(x) = h^n F^{(n)}(1, \ldots, 1; x + \tfrac{1}{2}nh, x + \tfrac{1}{2}nh - h, \ldots, x - \tfrac{1}{2}nh).$$

5.5-4 If $\mu, \nu \in \mathbb{C}_>$ and f is continuous, show that the fractional integral (5.5-15) satisfies $I^\mu(I^\nu f) = I^{\mu+\nu} f$.

5.5-5 Let Ω be an open interval in $\mathbb{R}$ and let the complex-valued functions f and g be continuous on Ω. Assume $a, x \in \Omega$ and $\beta, \beta' \in \mathbb{C}_>$. Let

$$h(x) = \frac{1}{\Gamma(\beta+\beta')} \int_a^x (x-t)^{\beta+\beta'-1} F(\beta, \beta'; x, t) g(t)\, dt.$$

Show that $h = I^{\beta'} f I^\beta g$, where the operator I is defined by (5.5-15). (This result is the basis of a method of finding g when h and f are given.)

5.5-6 If $\beta, \nu \in \mathbb{C}_>$, a generalized fractional integral of f is defined by

$$J^\nu f(x) = (x-a)^{\beta+\nu-1} \frac{\Gamma(\beta)}{\Gamma(\beta+\nu)} F(\beta, \nu; x, a).$$

If f is continuous, show that

$$\frac{d}{dx} J^{\nu+1} f(x) = J^{\nu} f(x).$$

5.6-1 Let Ω be either an open interval in $\mathbb{R}$ or a convex open set in $\mathbb{C}$, and let $f: \Omega \to \mathbb{C}$ be continuous. Define $h: \Omega \to \mathbb{C}$ by $h(x) = xf(x)$, and let H be the Dirichlet average of h. If w is defined by (5.6-2), show that

$$H(b, z) = \sum_{i=1}^{k} w_i z_i \, F(b + e_i, z).$$

5.6-2 Let Ω be either an open interval in $\mathbb{R}$ or a convex open set in $\mathbb{C}$, and let $f: \Omega \to \mathbb{C}$ be continuous. If $c = \sum_{i=1}^{k} b_i$ and $n \in \mathbb{N}$, show that

$$F(b, z) = \frac{n!}{(c, n)} \sum \frac{(b_1, m_1) \cdots (b_k, m_k)}{m_1! \cdots m_k!} F(b + m, z),$$

where the summation extends over all nonnegative integers $m_1, \ldots, m_k$ whose sum is n.

5.7-1 Let ρ be the radius of convergence of the hypergeometric series $g(x) = {}_pF_q[(a), (c), x]$. If $\operatorname{Re} \gamma > \operatorname{Re} \beta > 0$ and $|x| < \rho$, show that

$$G(\beta, \gamma - \beta; x, 0) = {}_{p+1}F_{q+1}[(a), \beta; (c), \gamma; x],$$

where $(a), \beta$ stands for the array $a_1, \ldots, a_p, \beta$.

5.8-1 Prove that (5.8-3) is equivalent to the second equation of (5.8-5).

5.8-2 If $x \in \mathbb{C}$ and $\operatorname{Re} \nu > \frac{1}{2}$, show that the Bessel function J_ν satisfies

$$2\nu J_\nu(x) = xJ_{\nu+1}(x) + xJ_{\nu-1}(x).$$

5.8-3 Let $p - 1 \in \mathbb{N}$ and $x \in \mathbb{C}$. Show that

$$e^x - \sum_{n=0}^{p-1} \frac{x^n}{n!} = \frac{x^p}{p!} S(1, p; x, 0).$$

5.8-4 Let $x \in \mathbb{R}^k$ and $b \in \mathbb{R}_>^{\,k}$, and define w by (5.6-2). Show that

$$\exp\left(\sum_{i=1}^{k} w_i x_i \right) \le S(b, x) \le \sum_{i=1}^{k} w_i e^{x_i}.$$

5.8-5 Assume $b \in \mathbb{R}_>^{\,k}$ and define the weights w by (5.6-2). Let the function f be defined and strictly positive on an interval $I \subset \mathbb{R}$. For every $x \in I^k$ define $\lambda_i(x) = \log f(x_i)$, $i = 1, \ldots, k$. If f is log-convex on I, prove that

$$F(b, x) \le S(b, \lambda(x)) \le \sum_{i=1}^{k} w_i f(x_i), \qquad x \in I^k.$$

If $1/f$ is log-convex on I, prove that

$$F(b, x) \ge S(b, \lambda(x)) \ge \prod_{i=1}^{k} [f(x_i)]^{w_i}, \qquad x \in I^k.$$

5.8-6 If $x, y \in \mathbb{C}$ show that

$$S(1, 1; x, y) = \frac{e^x - e^y}{x - y}, \qquad x \neq y,$$

$$\frac{\sin x}{x} = S(1, 1; ix, -ix), \qquad x \neq 0.$$

5.8-7 If x and μ are complex numbers with $\operatorname{Re}\mu > -\frac{1}{2}$, show that the Bessel function $J_\mu(x)$ can be represented by Poisson's integral,

$$\pi^{1/2}\Gamma(\tfrac{1}{2} + \mu)(x/2)^{-\mu}J_\mu(x) = \int_0^\pi e^{ix\cos\varphi}(\sin\varphi)^{2\mu}\, d\varphi = 2\int_0^{\pi/2} \cos(x\cos\varphi)(\sin\varphi)^{2\mu}\, d\varphi.$$

5.8-8 Let $\alpha, \delta, \sigma, x \in \mathbb{C}$ and $\beta \in \mathbb{C}_>$. Let f be holomorphic on a convex open set $\Omega \in \mathbb{C}$. If $\delta \pm \sigma x \in \Omega$, show that

$$\begin{aligned} &x^{1-\alpha}\frac{d}{dx}[x^\alpha F(\beta + 1, \beta + 1; \delta + \sigma x, \delta - \sigma x)] \\ &\quad = (2\beta + 1)F(\beta, \beta; \delta + \sigma x, \delta - \sigma x) + (\alpha - 2\beta - 1)F(\beta + 1, \beta + 1; \delta + \sigma x, \delta - \sigma x). \end{aligned}$$

In particular, if $\lambda \in \mathbb{C}$ and $\operatorname{Re}\nu > \frac{1}{2}$, show that Bessel's function satisfies

$$x^{1-\lambda}\frac{d}{dx}[x^\lambda J_\nu(x)] = xJ_{\nu-1}(x) + (\lambda - \nu)J_\nu(x).$$

5.8-9 If $x, \lambda \in \mathbb{C}$ and $\operatorname{Re}\nu > \frac{1}{2}$, show that Bessel's function satisfies

$$2\nu x^{-\lambda}\frac{d}{dx}[x^\lambda J_\nu(x)] = (\lambda + \nu)J_{\nu-1}(x) + (\lambda - \nu)J_{\nu+1}(x).$$

Note the cases $\lambda = \pm\nu$.

5.8-10 Let $n \in \mathbb{N}$ and $x, y \in \mathbb{C}$. Assume $x \neq y$. Show that

$$\left(\frac{e^x - e^y}{x - y}\right)^n = S[1, \ldots, 1; nx, (n-1)x + y, \ldots, x + (n-1)y, ny].$$

If $x \neq 0$, show that

$$\left(\frac{\sin x}{x}\right)^n = S[1, \ldots, 1; nix, (n-2)ix, \ldots, (-n+2)ix, -nix].$$

5.8-11 Let $\beta, \beta' \in \mathbb{C}_>$ and $\rho, \sigma, \tau \in \mathbb{C}$. Let $\varphi(x) = x^\rho S(\beta, \beta'; \sigma x, \tau x)$ and show that

$$x\left(\frac{d}{dx} - \sigma - \frac{\rho}{x}\right)\left(\frac{d}{dx} - \tau - \frac{\rho}{x}\right)\varphi + \beta\left(\frac{d}{dx} - \sigma - \frac{\rho}{x}\right)\varphi + \beta'\left(\frac{d}{dx} - \tau - \frac{\rho}{x}\right)\varphi = 0.$$

5.8-12 Let $k, x \in \mathbb{R}^n$, $n \geq 2$, and define $k \cdot x = \sum_{i=1}^n k_i x_i$ and $K = (\sum_{i=1}^n k_i^2)^{1/2}$. Assume that $f\colon \mathbb{R}_+ \to \mathbb{C}$ satisfies

$$\int_0^\infty |f(r)| r^{n-1}\, dr < \infty.$$

Show that

$$\int_{\mathbb{R}^n} e^{ik\cdot x} f\left[\left(\sum_{i=1}^{n} x_i^2\right)^{1/2}\right] dx_1 \cdots dx_n = \frac{2\pi^{n/2}}{\Gamma(n/2)} \int_0^\infty S\left(\frac{n-1}{2}, \frac{n-1}{2}; iKr, -iKr\right) f(r) r^{n-1}\, dr.$$

5.9-1 Let $p - 1 \in \mathbb{N}$, $\alpha \in \mathbb{C}$, and $1 - x \in \mathbb{C}_>$. Show that

$$(1-x)^{-\alpha} - \sum_{n=0}^{p-1} (\alpha, n)\frac{x^n}{n!} = (\alpha, p)\frac{x^p}{p!} R_{-\alpha-p}(1, p; 1-x, 1).$$

5.9-2 Let $\alpha \in \mathbb{C}$ and $\beta, \gamma - \beta, 1 - x \in \mathbb{C}_>$. Prove Euler's transformation,

$${}_2F_1(\alpha, \beta; \gamma; x) = (1-x)^{\gamma-\alpha-\beta}\, {}_2F_1(\gamma-\alpha, \gamma-\beta; \gamma; x).$$

5.9-3 Let $n \in \mathbb{N}$ and $\alpha, \beta, \gamma \in \mathbb{C}$. If $(\gamma, n) \neq 0$, show that

$$\sum_{m=0}^{n} \frac{(\alpha, m)(\beta, m)(\gamma-\alpha-\beta, n-m)}{(\gamma, m)m!\,(n-m)!} = \frac{(\gamma-\alpha, n)(\gamma-\beta, n)}{(\gamma, n)n!}.$$

Assume further that $(\gamma - \alpha - \beta, n) \neq 0$ and show that

$${}_3F_2(-n, \alpha, \beta; \gamma, 1-\gamma+\alpha+\beta-n; 1) = \frac{(\gamma-\alpha, n)(\gamma-\beta, n)}{(\gamma, n)(\gamma-\alpha-\beta, n)}.$$

This result, published by Pfaff in 1797 but commonly called Saalschütz' theorem, sums any terminating ${}_3F_2$ series with unit argument in which the sum of the denominator parameters exceeds by unity the sum of the numerator parameters.

5.9-4 If $x, y \in \mathbb{C}$ and $\beta, \gamma - \beta \in \mathbb{C}_>$, show that

$$e^{-xy}\, {}_1F_1(\beta; \gamma; y) = \sum_{n=0}^{\infty} {}_2F_1(-n, \beta; \gamma; 1/x)(-xy)^n/n!.$$

5.9-5 If $t \neq -1$, prove (5.9-6) from (5.9-7) and (5.9-5).

5.9-6 Let $b - e_i - e_j \in \mathbb{C}_>{}^k$ and define $c = \sum_{i=1}^k b_i$. Show that

$$t(z_i - z_j)R_{t-1}(b, z) + (c-1)[R_t(b-e_i, z) - R_t(b-e_j, z)] = 0,$$

$$(c+t-1)(z_i - z_j)R_t(b, z) + (c-1)[z_j R_t(b-e_i, z) - z_i R_t(b-e_j, z)] = 0.$$

The second relation is due to D. G. Zill, and the first is a special case of (5.6-10). A third relation comes from setting $F = R_t$ in (5.6-11).

5.9-7 If c and w are defined by (5.6-2), prove the relations

$$tR_{t-1}(b, z) = \sum_{i=1}^{k} w_i z_i^{-1}[(c+t)R_t(b+e_i, z) - cR_t(b, z)],$$

$$(t+1)\sum_{i=1}^{k} w_i z_i^2 R_t(b+e_i, z) = (c+t+1)R_{t+2}(b, z) - \sum_{i=1}^{k} b_i z_i R_{t+1}(b, z).$$

5.9-8 Prove the analog for $k = 3$ of (5.9-24):

$$tw_i(z_i - z_j)(z_i - z_k)R_{t-1}(b+e_i, z) = tz_j z_k R_{t-1}(b, z)$$
$$- \left[\sum_{s=1}^{3} b_s z_s + t(z_j + z_k)\right] R_t(b, z) + (c+t)R_{t+1}(b, z),$$

where $b = (b_1, b_2, b_3)$, $z = (z_1, z_2, z_3)$, and i, j, k is any permutation of 1, 2, 3.

5.9-9 Prove the analog for $k = 3$ of (5.9-25):

$$(c + t)(c + t + 1)R_{t+2}(b, z) - (c + t)\sum_{i=1}^{3}(t + 1 + b_i)z_i R_{t+1}(b, z)$$

$$+ (t + 1)\sum_{i=1}^{3}(c + t - b_i)z_j z_k R_t(b, z) - t(t + 1)z_1 z_2 z_3 R_{t-1}(b, z) = 0,$$

where $b = (b_1, b_2, b_3)$, $z = (z_1, z_2, z_3)$, and i, j, k is a permutation of 1, 2, 3.

5.9-10 For $k = 2$ show that

$$c(c + 1)R_t(b, z) = c(c + t + 1)R_t(b + e_1 + e_2, z) - t(b_1 z_2 + b_2 z_1)R_{t-1}(b + e_1 + e_2, z).$$

5.9-11 Assume $b, x \in \mathbb{R}_>^k$ and define the weights w by (5.6-2). Show that

$$\left(\sum_{i=1}^{k} w_i x_i\right)^t \leq R_t(b, x) \leq \sum_{i=1}^{k} w_i x_i^t$$

if $t \geq 1$ or $t \leq 0$, with reversed inequalities if $0 \leq t \leq 1$.

5.9-12 Let H be an open half-plane in $\mathbb{C} - \{0\}$. For every $z \in H^k$ and $b \in \mathbb{C}_>^k$, define

$$L(b, z) = \int_E \log(u \cdot z)\, d\mu_b(u).$$

Show that

$$L(b, \lambda z) = L(b, z) + \log \lambda, \qquad \lambda \in \mathbb{C} - \{0\},$$

$$-L(b, 1 - z) = \sum_{n=1}^{\infty} \frac{1}{n} R_n(b, z), \qquad \max_i |z_i| < 1,$$

$$L(b, z) = \left[\frac{\partial}{\partial t} R_t(b, z)\right]_{t=0},$$

$$\sum_{i=1}^{k} D_i L(b, z) = R_{-1}(b, z), \qquad \sum_{i=1}^{k} z_i D_i L(b, z) = 1,$$

$$L(1, 1; x, y) = \frac{x \log x - y \log y}{x - y} - 1, \qquad x \neq y.$$

Assume $b, x \in \mathbb{R}_>^k$ and define w by (5.6-2). Show that

$$\prod_{i=1}^{k} x_i^{w_i} \leq e^{L(b, x)} \leq \sum_{i=1}^{k} w_i x_i.$$

5.9-13 Let H be an open half-plane in $\mathbb{C} - \{0\}$. If $x, y \in H$ and $x \neq y$, show that

$$R_{t-1}(1, 1; x, y) = \frac{x^t - y^t}{t(x - y)}, \qquad t \neq 0,$$

$$R_{-1}(1, 1; x, y) = \frac{\log x - \log y}{x - y}.$$

5.9-14 The regular Coulomb wave function is a multiple of

$$\rho^{L+1}S(L+1-i\eta, L+1+i\eta; i\rho, -i\rho) = \sum_{k=L+1}^{\infty} A_k{}^L(\eta)\rho^k, \qquad L \in \mathbb{N}.$$

[Abramowitz (1964, Chapter 14) uses the function (5.8-22) multiplied by ρ.] In physical terms ρ, L, and η are respective dimensionless measures of the distance between two charged particles, their angular momentum, and the product of their charges divided by their relative speed. Show that

$$A_k{}^L(\eta) = \frac{1}{(k-L-1)!} R_{k-L-1}(L+1-i\eta, L+1+i\eta; i, -i),$$

and prove the recurrence relation

$$(k+L)(k-L-1)A_k{}^L = 2\eta A_{k-1}^L - A_{k-2}^L, \qquad k > L+2.$$

5.9-15 Do Exercise 4.2-10 by using the properties of the R function.

5.9-16 Let $z \in \mathbb{C}_>{}^2$, $b - e_1 - e_2 \in \mathbb{C}_>{}^2$, and $c = b_1 + b_2$. Show that

$$R_{t+1}(b - e_1 - e_2, z) - \frac{1}{2}\left[\frac{(2t+c)}{c(c-2)}(b_1 - b_2)(z_1 - z_2) + z_1 + z_2\right]R_t(b, z)$$
$$- \frac{t(t+c)}{c^2(c^2-1)} b_1 b_2 (z_1 - z_2)^2 R_{t-1}(b + e_1 + e_2, z) = 0.$$

5.9-17 If $x, y, \beta - 1 \in \mathbb{C}_>$, show that

$$t(t-1)(x-y)^2 R_{t-2}(\beta+1, \beta+1; x, y) = 4(4\beta^2 - 1)[R_t(\beta-1, \beta-1; x, y) - R_t(\beta, \beta; x, y)].$$

5.9-18 Let $n \in \mathbb{N}$, $a \in \mathbb{C}$, $b \in \mathbb{C}_>{}^k$. Let $x + z = (x + z_1, \ldots, x + z_k) \in \mathbb{C}_>{}^k$ and define $f(x) = R_{-a}(b, x + z)$. Show that $f^{(n)}(x) = (-1)^n(a, n)R_{-a-n}(b, x + z)$.

5.9-19 Let $t \in \mathbb{C}$, $b \in \mathbb{C}_>{}^2$, $z \in \mathbb{C}_>{}^2$. Show that

$$\int_0^{\pi/2} (z_1 \sin^2\theta + z_2 \cos^2\theta)^t (\sin\theta)^{2b_1-1}(\cos\theta)^{2b_2-1}\,d\theta = \tfrac{1}{2}B(b)R_t(b, z).$$

5.9-20 Let $t \in \mathbb{C}$, $b \in \mathbb{C}_>{}^3$, $z \in \mathbb{C}_>{}^3$, $(l, m, n) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. Show that

$$\int_0^{\pi/2}\int_0^{\pi/2} (z_1 l^2 + z_2 m^2 + z_3 n^2)^t l^{2b_1-1} m^{2b_2-1} n^{2b_3-1} \sin\theta\,d\theta\,d\varphi = \tfrac{1}{4}B(b)R_t(b, z).$$

5.9-21 Let $t + \frac{3}{2} \in \mathbb{C}_>$ and $\alpha, \beta, \gamma, \lambda, \mu, \nu \in \mathbb{R}_>$. Show that

$$\int_{\mathbb{R}^3} (\alpha x^2 + \beta y^2 + \gamma z^2)^t \exp(-\lambda x^2 - \mu y^2 - \nu z^2)\,dx\,dy\,dz$$
$$= 2\pi\Gamma(t + \tfrac{3}{2})(\lambda\mu\nu)^{-1/2} R_t(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; \alpha/\lambda, \beta/\mu, \gamma/\nu).$$

5.9-22 Let A and B be real positive definite matrices with k rows and k columns, and let $\lambda_1, \ldots, \lambda_k$ be the eigenvalues of AB^{-1}. Denote by $\tilde{x}Ax$ the quadratic form $\sum_{i=1}^k \sum_{j=1}^k A_{ij}x_i x_j$. Let $t \in \mathbb{C}$, and assume $\operatorname{Re} t > -k/2$. Show that

$$\int_{\mathbb{R}^k} (\tilde{x}Ax)^t \exp(-\tilde{x}Bx)\,dx_1 \cdots dx_k = \frac{\pi^{k/2}\Gamma(t + k/2)}{(\det B)^{1/2}\Gamma(k/2)} R_t\left(\frac{1}{2}, \ldots, \frac{1}{2}; \lambda_1, \ldots, \lambda_k\right).$$

5.9-23 Let $n \in \mathbb{N}$. Evaluate the following sum of $n+1$ terms:

$$1 - \frac{2^2n^2}{1\cdot 3} + \frac{2^4n^2(n^2-1^2)}{1\cdot 3\cdot 3\cdot 5} - \frac{2^6n^2(n^2-1^2)(n^2-2^2)}{1\cdot 3\cdot 3\cdot 5\cdot 5\cdot 7} + \cdots.$$

5.9-24 Let $x, y, z \in \mathbb{C}_>$. Show that

$$-\frac{1}{2}R_{-2}(1, 1, 1; x, y, z) = \frac{\log x}{(x-y)(x-z)} + \frac{\log y}{(y-z)(y-x)} + \frac{\log z}{(z-x)(z-y)}.$$

More generally, assume k, n, $k-n-2 \in \mathbb{N}$ and $z \in \mathbb{C}_>{}^k$. Define $p_k(t) = (t-z_1)\cdots(t-z_k)$ and show that

$$R_{n-k+1}(1, \ldots, 1; z_1, \ldots, z_k) = (-1)^{k-1}\frac{(1-k, n+1)}{n!}\sum_{i=1}^{k}\frac{z_i^n \log z_i}{p_k'(z_i)}.$$

5.9-25 Let $t, x, y \in \mathbb{C}$ and $\varphi \in \mathbb{R}$. Assume (unless $t \in \mathbb{N}$) that 0 is not on the line segment with endpoints $x+y$ and $x-y$. Show that

$$\frac{1}{2\pi}\int_0^{2\pi}[x + y\cos(\theta+\varphi)]^t\,d\theta = R_t\left(\frac{1}{2}, \frac{1}{2}; x+y, x-y\right).$$

5.10-1 Assume $x \in \mathbb{C}$, $a \in \mathbb{C}^p$, $\alpha \in \mathbb{C}$, $c \in U^q$, and $p \le q$. Let $(a), \alpha$ stand for the array $a_1, \ldots, a_p, \alpha$. Prove that

$$\lim_{\alpha\to\infty} {}_{p+1}F_q[(a), \alpha; (c); x/\alpha] = {}_pF_q[(a); (c); x].$$

The series on the right is called a confluent limit of the series on the left, and the case $p = q = 0$ is (5.10-1).

5.10-2 Prove (5.10-8) and (5.10-9).

5.10-3 Let $x \in \mathbb{C}$ and $p, \lambda, \nu + \frac{1}{2} \in \mathbb{C}_>$. If $\operatorname{Re} p > |\operatorname{Im} x|$, show that

$$\int_0^\infty e^{-pt}t^{\lambda-\nu-1}J_\nu(xt)\,dt = \frac{\Gamma(\lambda)}{\Gamma(\nu+1)}\left(\frac{x}{2}\right)^\nu R_{-\lambda}\left(\nu+\frac{1}{2}, \nu+\frac{1}{2}; p+ix, p-ix\right).$$

5.10-4 Let $x \in \mathbb{C}$ and $p, \nu + \frac{1}{2} \in \mathbb{C}_>$. If $\operatorname{Re} p > |\operatorname{Im} x|$, show that

$$\int_0^\infty e^{-pt}t^\nu J_\nu(xt)\,dt = 2^\nu\pi^{-1/2}\Gamma(\nu+\tfrac{1}{2})x^\nu(p^2+x^2)^{-\nu-1/2}.$$

5.10-5 Let $n \in \mathbb{N}$, $x \in \mathbb{R}$, and $a, p \in \mathbb{C}_>$. Show that

$$\int_0^\infty e^{-pt}t^{a-n-1}\sin^n xt\,dt$$
$$= \Gamma(a)x^nR_{-a}(1, \ldots, 1; p+inx, p+inx-i2x, \ldots, p-inx+i2x, p-inx).$$

5.11-1 Show that the Bessel functions of orders $0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ have the representation

$$J_\nu(x) = \frac{(x/2)^\nu(2\nu)!}{\Gamma(1+\nu)2\pi i}\int_\gamma e^s(s^2+x^2)^{-\nu-1/2}\,ds,$$

where $2\nu \in \mathbb{N}$, $x \in \mathbb{C}$, and γ is a closed contour encircling in the positive direction the line segment with endpoints ix and $-ix$.

5.12-1 Let $x \in \mathbb{R}_>$. The error function and complementary error function are defined by

$$\operatorname{Erf}(x) = \int_0^x e^{-t^2}\,dt, \qquad \operatorname{Erfc}(x) = \int_x^\infty e^{-t^2}\,dt.$$

Show that

$$\operatorname{Erf}(x) = xS(\tfrac{1}{2}, 1; -x^2, 0), \qquad \operatorname{Erfc}(x) = \tfrac{1}{2}e^{-x^2}x^{-1}\,{}_2F_0(\tfrac{1}{2}, 1; -1/x^2).$$

5.12-2 Let $a \in \mathbb{C}_>$ and $x \in \mathbb{R}_>$. The incomplete gamma functions are defined by

$$\gamma(a, x) = \int_0^x t^{a-1}e^{-t}\,dt, \qquad \Gamma(a, x) = \int_x^\infty t^{a-1}e^{-t}\,dt.$$

Show that

$$\gamma(a, x) = a^{-1}x^aS(a, 1; -x, 0), \qquad \Gamma(a, x) = e^{-x}x^{a-1}\,{}_2F_0(1 - a; 1; -1/x).$$

5.12-3 Let $-x \in \mathbb{R}_>$. The exponential integral is defined by

$$\operatorname{Ei}(x) = \int_{-\infty}^x e^t t^{-1}\,dt.$$

Show that

$$\operatorname{Ei}(x) = e^x x^{-1}\,{}_2F_0(1, 1; 1/x).$$

5.12-4 Let $0 \le x < 1$. The logarithmic integral is defined by

$$\operatorname{li}(x) = \int_0^x \frac{dt}{\log t}.$$

Show that

$$\operatorname{li}(x) = x(\log x)^{-1}\,{}_2F_0(1, 1; 1/\log x).$$